FSB FINANCIAL STABILITY BOARD

FSB Global Regulatory Framework for Crypto-Asset Activities

Umbrella public note to accompany final framework



17 July 2023

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Overview

The FSB is finalising its global regulatory framework for crypto-asset activities to promote the comprehensiveness and international consistency of regulatory and supervisory approaches. It consists of two distinct sets of recommendations:

(i) High-level recommendations for the regulation, supervision and oversight of crypto-asset activities and markets (CA recommendations);

(ii) Revised high-level recommendations for the regulation, supervision, and oversight of "global stablecoin" arrangements (GSC recommendations).

The framework is based on the principle of "same activity, same risk, same regulation" and provides a strong basis for ensuring that crypto-asset activities and so-called stablecoins are subject to consistent and comprehensive regulation, commensurate to the risks they pose, while supporting responsible innovations potentially brought by the technological change. The recommendations focus on addressing risks to financial stability, and they do not comprehensively cover all specific risk categories related to crypto-asset activities.[1] It takes account of lessons from recent events in crypto-asset markets. Central Bank Digital Currencies (CBDCs), envisaged as digitalised central bank liabilities, are not subject to these recommendations.

The events of the past year have highlighted the intrinsic volatility and structural vulnerabilities of crypto-assets and related players. They have also illustrated that the failure of a key service provider in the crypto-asset ecosystem can quickly transmit risks to other parts of that ecosystem. As recent events have illustrated, if linkages to traditional finance were to grow further, spillovers from crypto-asset markets into the broader financial system could increase. The G20 has asked the FSB to coordinate the delivery of an effective regulatory, supervisory and oversight framework for crypto-assets[2], including finalising the FSB's high-level recommendations for the supervision and regulation of crypto-asset activities, and of so-called global stablecoins (GSCs), by July 2023. In addition, these recommendations, constituting a regulatory and supervisory framework for crypto-assets and stablecoins, will provide input to a joint paper with the International Monetary Fund (IMF) to be delivered to the G20 in September 2023, which will support a coordinated and comprehensive policy approach to crypto-assets by synthesising the policy findings from IMF work on macroeconomic and monetary issues and FSB work on supervisory and regulatory issues.

The FSB and the sectoral standard-setting bodies (SSBs) have developed a shared workplan for 2023 and beyond, through which they will continue to coordinate work under their respective mandates to promote the development of a comprehensive and coherent global regulatory framework commensurate to the risks crypto-asset markets activities may pose to jurisdictions

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1 Other risks include AML/CFT; data privacy; cyber security; consumer and investor protection; market integrity; competition policy; taxation; monetary policy; monetary sovereignty and other macroeconomic concerns.

2 *Communiqué G20 Finance Ministers and Central Bank Governors Meeting*, 17-18 February, Jakarta, Indonesia.

worldwide, including through the provision of more granular guidance by SSBs, monitoring and public reporting.

1. Background

Crypto-asset activities are global in nature, and these various activities transcend sectoral lines that divide traditional financial activities. The FSB, with its core mandate to coordinate the development of regulatory, supervisory, and other financial sector policies, was tasked by the G20 to promote cooperation among jurisdictional financial authorities and SSBs to ensure that crypto-asset activities are subject to robust regulation and supervision commensurate to the financial stability risks that they pose, while supporting responsible innovation.[3]

In October 2022, the FSB published a proposed version of this framework, in the form of two separate sets of recommendations, for a public consultation that ran until 15 December 2022. A summary of the consultation responses, together with the individual responses, has been published separately. The FSB has carefully considered the responses, and held an outreach meeting in February 2023 to further discuss the responses from different viewpoints[4]. In light of that consideration and a review of recent events in crypto-asset markets, the FSB is now publishing the final versions of the recommendations.

This document (i) describes how the two distinct sets of recommendations build a framework for the regulation, supervision and oversight of global stablecoins arrangements and other crypto-asset activities, that also addresses lessons from the events in 2022 and early 2023 in crypto-asset markets; and (ii) sets out next steps.

The FSB has been working closely with the Basel Committee on Banking Supervision (BCBS), Bank for International Settlements' Committee on Payments and Market Infrastructures (CPMI), International Organization of Securities Commissions (IOSCO), Financial Action Task Force (FATF), IMF, World Bank, and Organization for Economic Cooperation and Development (OECD) to ensure that the work underway regarding the monitoring and regulation of crypto-asset activities and markets is coordinated, mutually supportive, and complementary.

The FSB high-level recommendations take a broad approach to global stablecoins. International standards designed for specific sectors focus on distinct functions within the remit of the relevant standard setting bodies (SSBs). Where international sectoral standards apply to a GSC for a particular economic function, those standards will address risks specific to the economic function and, as such, authorities should implement those international standards.

To provide a holistic view of the crypto-asset regulatory and supervisory policy initiatives, the FSB and SSBs have developed a shared workplan that brings together their completed, ongoing,

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3 *Communiqué G20 Finance Ministers and Central Bank Governors Meeting:* "We encourage the FSB, in close coordination with other standard-setting bodies, to accelerate and deepen its work to monitor and share information on regulatory and supervisory approaches to unbacked cryptoassets, stablecoins, decentralized finance, and other forms of crypto-assets and to address any gaps and arbitrage, including by recommending coordinated and timely policy actions to preserve global financial stability, thus creating the necessary conditions for safe innovation." G20, 17-18 February 2022, Jakarta, Indonesia.

4 Comments on the consistency between both sets of recommendations have been addressed. In several cases differences in wording between the two sets of recommendations exist to account for differences between stablecoins and crypto-assets.

and planned work. The SSBs are taking forward work, either individually or jointly, to help regulators tackle risks arising from crypto-assets and GSCs within their mandates and avoid over- and under-laps in the regulation of the sector. The SSBs' work is critical in in taking forward more detailed sectoral standard setting, where needed, and exploring issues unique to each SSB's mandate that may necessitate additional analytical work or policy measures in due course. A table reflecting a preliminary round of inputs by the FSB and SSBs is included in the Annex to this document.

2. The FSB's approach to establishing a comprehensive regulatory framework

Since the publication of the FSB consultative reports, jurisdictional authorities have continued to work to strengthen the regulation and supervision of crypto-asset activities. Several jurisdictions have made progress in developing regulatory frameworks for crypto-assets and stablecoins and related activities. In some other jurisdictions, authorities continue to apply and enforce existing applicable requirements, laws and regulations on crypto-asset market participants that had been conducting activities in non-compliance with these existing requirements, laws and regulations. The development of regulations and implementation of requirements, and application and enforcement of existing requirements are consistent with the FSB's framework and demonstrate that crypto-asset activities do not operate in a regulation-free space. Crypto-asset service providers must not commence operations in any jurisdiction unless any such service provider meets all applicable regulatory requirements.

Taking into account the responses to the public consultation, the final recommendations continue to build on the principles that informed the consultative framework, while integrating key aspects drawing on the experiences of the implementation of international standards by jurisdictional authorities. These principles are:

- "Same activity, same risk, same regulation"

  The FSB recommendations aim at ensuring that regulatory frameworks are applied to global stablecoin and crypto-asset activities commensurate to the risks that they may pose to financial stability. Regulatory frameworks should also ensure that any crypto-asset activities seeking to perform an equivalent economic function to one in the traditional financial system are subject to the same or equivalent regulation irrespective of the way crypto-asset activities are conducted or the way these activities are marketed.

- High-level and flexible

  The FSB recommendations are high-level and allow for sufficient flexibility for jurisdictional authorities to implement them by applying relevant current regulations or developing new domestic regulatory frameworks, and to adapt to a currently rapidly evolving market. This approach also leaves adequate room for SSBs to develop granular standards that address sectoral issues within their respective mandates.

- Technology neutral

  The FSB recommendations focus on addressing financial stability risks associated with global stablecoin and crypto-asset activities and are designed to be applicable across the different technologies that may be employed. Crypto-asset activities must be regulated based on their economic functions and the risks they pose, irrespective of the technological means used.

While the CA recommendations apply to any type of crypto-asset activity, including stablecoins, those stablecoins that could be widely used as a means of payment and/or store of value across multiple jurisdictions – GSCs – could pose particular risks to financial stability. Therefore, separate and complementary recommendations for crypto-asset activities that meet the definition of a GSC have been published to reflect their particular risks and heightened regulatory and supervisory requirements. The table below clarifies which regulatory principle is covered in each of the two sets of recommendations. In response to the public consultation, the language in each set of recommendations has been made more consistent with the other.

Table 1: Coverage of the CA and GSC Recommendations

Regulatory principle	Coverage in GSC recommendations	Coverage in CA recommendations
Regulatory powers	GSC Rec 1	CA Rec 1
Comprehensive oversight	GSC Rec 2	CA Rec 2
Cross-border cooperation	GSC Rec 3 with an Annex	CA Rec 3
Governance	GSC Rec 4	CA Rec 4
Risk management	GSC Rec 5	CA Rec 5
Data management	GSC Rec 6	CA Rec 6
Recovery and resolution planning	GSC Rec 7	CA Rec 5
Disclosures	GSC Rec 8 with an Annex	CA Rec 7
Monitoring of interconnections within the crypto-asset ecosystem and with the wider financial system	[N/A]	CA Rec 8
Compliance before operation	GSC Rec 10	CA Rec 1
Redemption rights and stabilisation mechanisms	GSC Rec 9	[N/A]
Multiple functions	[N/A]	CA Rec 9

3. Recent events and implications for the regulation and supervision of crypto-asset activities and stablecoins

The crypto-asset market has experienced significant turmoil since early 2022. In May 2022, a sharp decrease of crypto-asset prices demonstrated that the inherently high volatility of crypto-asset markets can lead to sudden market strains and failures of crypto-asset intermediaries. The demise of the TerraUSD/LUNA coins and the de-pegging of a number of so-called stablecoins also highlighted their ineffective stabilisation mechanisms. The crypto-asset market was shaken again in November when FTX, one of the largest crypto-asset trading platforms at that time, went into bankruptcy. The FTX event has revealed a number of vulnerabilities that may arise from the vertical integration of business activities that are not adequately regulated in all jurisdictions in which they operate, or are in non-compliance with existing regulations, as well as risks arising from non-compliance with existing requirements, including: a lack of transparency of corporate structure, key function holders and financial positions; misappropriation of clients' funds; reliance on self-issued crypto-assets; and interconnectedness with affiliated entities. More recently in the US, the decline in crypto-asset prices and activity led to one bank that offered services to crypto-asset clients to close and may have contributed to the failure of another bank. Conversely, the failure of a regional bank that offered deposit services to a stablecoin issuer contributed to a temporary de-pegging of its stablecoin because of uncertainty about the issuer's access to its deposits.

These events demonstrate interlinkages between crypto-asset markets and the traditional financial system. Although spillovers to the traditional financial system have been limited, stress events in crypto-asset markets caused significant losses to investors and shook confidence in these markets. Given the speed of developments in this area, interlinkages and possible spillovers between the crypto-asset and traditional financial systems could markedly increase, reinforcing the need to be forward-looking and the importance of ensuring that the risks posed by crypto-assets to financial stability are addressed by appropriate and comprehensive regulation and supervision.

These events have further highlighted that many crypto-asset activities involve economic functions similar to those in traditional finance and should be comprehensively regulated and/or brought into compliance with new or existing rules as appropriate. Most failed market participants, including those behind TerraUSD/LUNA, Celsius Network, and FTX, engaged in functions similar to those of traditional finance. Those functions, when carried out without appropriate governance structures or in non-compliance with regulatory requirements, created vulnerabilities that have been identified by the FSB. They include liquidity and maturity mismatch, the build-up of leverage, the creation of complex interrelationships between service providers, and the integration of multiple functions under one or more related entities without appropriate governance structures, regulatory oversight, or in non-compliance with applicable regulatory frameworks. In light of these lessons learned, the FSB has strengthened both sets of high-level recommendations in three areas:

- **Safeguarding of client assets**: Financial service providers that hold or take control of assets from clients must ensure that these assets are effectively segregated from their own assets. Many crypto-asset service providers, however, are observed to extensively commingle proprietary assets with client assets, often without consent of the clients. This enables these service providers to misuse customer assets and to amplify leverage

and liquidity transformation by the re-use of the commingled assets. When some of these intermediaries failed, significant losses to clients and investors resulted, and the failure to clarify ownership rights added complexity to insolvency proceedings. The high-level recommendations have been strengthened by stating that authorities should require that crypto-asset service providers maintain adequate safeguarding of customer assets and protect ownership rights, including in insolvency.

- **Conflicts of interest**: Various crypto-asset intermediaries combine multiple functions. In some cases, they operate in ways that do not comply with existing requirements, which typically require that certain functions be conducted by different entities and subject to specific sectoral standards. Some entities are not transparent about their governance structures and set up complex structures of affiliated entities that often finance each other, leading to acute conflicts of interest and increasing interconnectedness and the risk of contagion within crypto-asset markets. The FSB has strengthened the high-level recommendations by stating that authorities should have in place requirements to address the risks associated with conflicts of interest. Crypto-asset service providers, including their affiliates, that combine multiple functions and activities must be subject to, or should be brought into compliance with, appropriate regulation, supervision and oversight, including, as appropriate, legal separation of certain functions.

- **Cross-border cooperation**: The borderless nature of crypto-assets underlines the importance of strong and consistent jurisdictional regulatory, supervisory and enforcement practices. Crypto-asset issuers and service providers may seek to evade regulation and oversight by migrating to places where regulation is lighter. This requires strong measures to address regulatory arbitrage. The FSB's high-level recommendations have been strengthened on information sharing, including about the level of compliance of activities spanning multiple jurisdictions, especially those in jurisdictions that have not implemented international standards.

4. Next steps

The FSB and the IMF will deliver a joint report to the G20 in September 2023, which will synthesise the policy findings from IMF work on macroeconomic and monetary issues (as set out in its February 2023 note to the G20 on Macrofinancial Implications of Crypto Assets[5]) and FSB work on supervisory and regulatory issues published today. This will support a coordinated and comprehensive policy approach to crypto-assets by considering macroeconomic and regulatory perspectives, including the full range of risks posed by crypto-assets.

As an important part of this approach, the FSB and the SSBs will continue to work on enhancing effectiveness and international consistency of the regulation, supervision and oversight of crypto-asset and stablecoin activities. While the high-level recommendations apply to so-called decentralised finance and service providers that combine multiple functions, the FSB will also conduct follow-up policy work, as appropriate, on these areas to determine if additional guidance or recommendations are necessary. The FSB will also explore ways to address the cross-border

5 IMF (2023), *G20 Note on the Macrofinancial Implications of Crypto Assets*, February.

risks GSCs pose to emerging market and developing economies and consider ways to strengthen supervisory and regulatory coordination.

The SSBs will, within their respective mandates continue to examine how existing standards apply to crypto-assets and stablecoin arrangements, and make revisions, as needed, to their sectoral international standards and guidance or, if necessary, provide additional guidance that supplements existing standards and principles in light of the FSB recommendations. Any additional granular guidance from SSBs will further articulate regulatory expectations at the global level, contributing to consistent outcomes under their respective mandates while mitigating the risk of regulatory arbitrage.

As many crypto-asset activities are still operating without being subject to comprehensive regulation, or being undertaken in non-compliance with applicable jurisdictional regulations, it is important that FSB members implement these high-level recommendations and relevant international standards fully and consistently. Jurisdictional authorities may apply and enforce existing requirements and laws, as well as develop new or additional regulatory approaches. The FSB will continue to facilitate discussions and information sharing amongst FSB members where such coordination supports an effective collective supervisory response to developments in crypto-asset markets. FSB will also regularly assess the need for further international policy work on crypto-asset activities, such as ways to support jurisdictional implementation.

The FSB will, by end-2025, conduct a review of the status of the implementation of these two sets of recommendations at the jurisdictional level. Considering that many crypto-asset activities are taking place, licensed or registered in non-FSB member jurisdictions, the FSB will, alongside the SSBs and other international agencies, take steps to promote and monitor the effective implementation in jurisdictions beyond the FSB's membership, including engaging with jurisdictions with material crypto-asset activities to reduce the scope for regulatory arbitrage. As part of this work, the FSB, together with the IMF, the Work Bank, and the SSBs, will

- engage with a wider set of jurisdictions to encourage implementation of the FSB recommendations and international standards,
- take stock of regulatory progress and challenges among non-FSB member jurisdictions through the FSB's Regional Consultative Groups, and
- monitor and address cross-border issues related to stablecoins and other crypto-assets.

5. Annex: FSB-SSB shared workplan

Initiative (Responsible Body)	Objective/Scope	Status/Timeline
Risks and regulation of crypto-assets and markets		
Finalisation of FSB high-level recommendations for regulation, supervision and oversight of crypto-assets and markets (FSB)	Recommendations to address financial stability risks of crypto-asset markets and activities, including issuers and service providers. They cover regulatory power, cross-border cooperation, governance, risk management, data, disclosure, systemic risk and multi-function intermediaries.	Publication of finalised high-level recommendations in July 2023 (G20), review of the status of implementation at a jurisdictional level by end-2025.
Crypto-Assets and Digital Assets Recommendations (IOSCO)	Apply IOSCO Principles and Standards to economically equivalent crypto-assets and activities to address the sizeable and proximate market integrity and investor protection risks in the sector, covering conflicts of interest, client asset protection, market manipulation, operational risk, retail access suitability, and cross-border issues, among others. Together with the IOSCO DeFi Recommendations and Guidance (see below), this will promote greater consistency with respect to how IOSCO members approach the regulation and oversight of crypto-asset activities, given the cross-border nature of the markets, the risks of regulatory arbitrage and the significant risk of harm to which retail investors continue to be exposed.	IOSCO Fintech Task Force (FTF) issued a consultation on proposed CDA policy recommendations in May 2023, focusing on crypto-asset service providers (CASPs). Following the consultation period, IOSCO will review feedback and finalise policy recommendations to address investor protection and market integrity risks in crypto-asset markets. The CDA policy recommendations will be finalised by end of -2023.

Initiative (Responsible Body)	Objective/Scope	Status/Timeline
Financial stability risks of decentralised finance (FSB)	To assess financial stability implications of DeFi (and associated trading, lending, and borrowing practices, services and platforms, protocols, and market participants) and draw policy implications that SRC could consider.	Report published and submitted to G20FMCBG in February.
Policy implications of DeFi (FSB)	To consider regulatory implications of DeFi and assess whether additional policy work is warranted.	Preliminary work underway and is expected to be completed by end-2024.
DeFi Recommendations (IOSCO)	Updates on the state of the DeFi market and provide recommendations on applying existing IOSCO principles to DeFi by mapping them against key aspects of DeFi activities and protocols, covering emerging issues, risks and considerations.	The IOSCO FTF, through its DeFi workstream, will consult on proposed DeFi policy recommendations during Q3 2023. Following the consultation period, IOSCO will review feedback and finalise policy recommendations to address investor protection and market integrity risks in DeFi. The DeFi policy recommendations will be finalised by end-2023.
Exploratory work on DeFi in the insurance sector (IAIS)	To explore recent developments in the area of Decentralised Insurance, analyse supervisory challenges and identify areas of future work.	Following initial analysis during 2021-2022, the IAIS Fintech Forum will continue analysis of developments related to decentralized insurance business models. Non-public note to be finalised in 2023. FSI-IAIS-EIOPA joint webinar in October 2023.
Multifunction crypto-asset intermediaries (FSB)	To assess financial stability implications of multifunction crypto-asset intermediaries, building on the lessons learnt from recent failures of such intermediaries, and draw policy implications that the FSB could consider.	Work underway.
Policy implications of multi-function crypto-asset intermediaries (FSB)	Consider regulatory implications and assess whether additional policy work is warranted arising from multifunction crypto-asset intermediaries that may give rise to compounded risks, conflict of interests, and opaque transactions with related parties that may not typically be allowed in traditional financial companies	Work to begin depending on the outcome of the FSB's analytic work and will be completed by end- 2024.

Initiative (Responsible Body)	Objective/Scope	Status/Timeline
Regular monitoring and assessment of vulnerabilities in crypto-asset markets (FSB)	To keep track of crypto-asset market developments (including stablecoins and DeFi) and assess their financial stability implications and conduct on-the-spot analysis of major crypto-asset market events.	Work underway.
Risks and regulation of Stablecoins		
Finalisation of FSB high-level recommendations for global stablecoins (GSCs) (FSB)	The FSB's recommendations for GSCs seek to promote consistent and effective regulation, supervision and oversight of GSCs across jurisdictions to address the potential financial stability risks posed by GSCs.	Publication of finalised high-level recommendations in July 2023 (G20), review of the status of implementation at a jurisdictional level by end-2025.
Authorities' responsibilities for stablecoin arrangements under the Principles for financial market infrastructure (PFMI) (CPMI-IOSCO)	To analyse challenges for relevant regulatory, supervisory and oversight authorities in implementing their PFMI Responsibilities for stablecoin arrangements, especially on Responsibility E (cooperation with other authorities) and identify potential approaches to addressing these challenges.	As appropriate, the key findings of the report will be shared with the wider international regulatory community
Risks associated with multi-currency and asset-linked stablecoins (CPMI-IOSCO)	To identify issues for multicurrency or asset-linked stablecoins that may require additional clarifications or additions to the July 2022 CPMI-IOSCO guidance on the application of the PFMI to SAs, including notable features of such stablecoin arrangements that might affect their compliance with the PFMI.	Publication on issues potentially requiring further clarifications related to multicurrency stablecoins used as a settlement asset is envisaged for H2 2023. The work on the asset-linked stablecoins is in the early stage
Cross-border regulatory coordination for GSCs in emerging market and developing economies (EMDEs) (FSB)	To explore how to address the unique cross-border financial stability risks posed by GSCs, including risks specific to EMDEs, and consider ways to enhance supervisory and regulatory coordination.	Work to begin in late Q3 2023 and continue into 2024.

Initiative (Responsible Body)	Objective/Scope	Status/Timeline
Banks' stablecoin exposures and related activities (BCBS)	See related BCBS activities in the section on Financial sector exposures to crypto-assets; in particular: criteria to identify stablecoins eligible for "Group 1b" prudential treatment, the composition of stablecoin reserve assets and banks as stablecoin issuers.	Ongoing as part of the BCBS 2023-24 work programme.
Regular monitoring of developments in stablecoins (CPMI secretariat)	To monitor the key trends in the stablecoin market, e.g., market capitalisation, deviations from peg and composition of backing assets.	Data being collected on an ongoing basis to support the work of the CPMI, BIS and the FSB.
Potential of SAs for cross-border payments as part of Building block 18 in the cross-border payments programme (CPMI)	To help central banks and regulatory authorities to better understand how SAs, if properly designed and regulated, could contribute to the objective of making cross-border payments faster, cheaper, more transparent and more inclusive. To analyse potential benefits and challenges from cross-border use of SAs, as well as potential implications for central banks' key functions.	The report is envisaged for publication in September 2023.
Financial sector exposures to crypto-assets		
Banks' exposures to crypto-assets (BCBS)	Monitor the implementation of BCBS's prudential treatment of banks' crypto-asset exposures. Review by end-2023 the treatment of permissionless blockchains with additional safeguards and the criteria to identify stablecoins eligible for the "Group 1b" prudential treatment, including the appropriate composition of reserve assets and the effectiveness of statistical tests.	Ongoing as part of the BCBS 2023-24 work programme.

Initiative (Responsible Body)	Objective/Scope	Status/Timeline
Assessment of bank-related developments and activities in crypto-asset markets (BCBS)	To assess bank-related developments and activities in crypto-asset markets, - including the role of banks as stablecoin issuers, custodians of crypto-assets and broader potential channels of interconnections with the crypto-asset ecosystem – and develop proposals to address any risks that are not adequately captured.	Ongoing as part of the BCBS 2023-24 work programme.
Monitoring of insurer-related developments in crypto-asset markets (IAIS)	To monitor the extent to which developments in crypto-asset markets may impact the insurance sector.	The IAIS Fintech Forum and IAIS Macroprudential Supervision Working Group undertook initial monitoring during 2022 – and concluded that the usage of and exposure to crypto-assets in the insurance sector is currently very limited. Further monitoring is planned for 2023. Next Fintech Forum roundtable on crypto-assets is planned for H2 2023.
Implications of distributed ledger technology (DLT) and DeFi for financial market infrastructures (FMIs) (CPMI-IOSCO)	To identify emerging risks associated with the use of DLT by FMIs and interaction between FMIs and Defi.	A workshop with industry and academics is envisaged for H2 2023.
Other areas of related work		
Tokenisation (FSB)	To take stock of recent developments, including ongoing or planned projects, regarding tokenisation of assets with the aim of identifying potential financial stability implications and exploring policy implications that warrant further consideration by the SCAV, SRC and other relevant bodies.	Work is underway, expected completion in 2024.

Initiative (Responsible Body)	Objective/Scope	Status/Timeline
Future of Payments - Tokenisation (CPMI)	Analytical exploration of the characteristics of tokenisation of money and payments, and the benefits, risks and challenges to central banks of a tokenised financial ecosystem	The work is at an early stage. Deliverables to be determined.
Annual CBDC and crypto-asset survey (CPMI)	To monitor the central banks' involvement in CBDC work as well as their perceptions regarding usage of stablecoins and other crypto-assets in their jurisdictions.	6th survey conducted with responses from over 80 central banks. Published in July 2023.]

FSB FINANCIAL STABILITY BOARD

High-level Recommendations for the Regulation, Supervision and Oversight of Crypto-Asset Activities and Markets

Final report



17 July 2023

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1. Introduction

Crypto-assets, as defined by the FSB[1], are a type of private sector digital asset that depend primarily on cryptography and distributed ledger or similar technology. The FSB in its crypto-assets report published in February 2022 concluded that "crypto-assets markets are fast evolving and could reach a point where they represent a threat to global financial stability".

The February 2022 G20 Finance Ministers and Central Bank Governors Communiqué requested:

"*We encourage the FSB, in close coordination with other standard-setting bodies, to accelerate and deepen its work to monitor and share information on regulatory and supervisory approaches to unbacked crypto-assets, stablecoins, decentralized finance, and other forms of crypto-assets and to address any gaps and arbitrage, including by recommending coordinated and timely policy actions to preserve global financial stability, thus creating the necessary conditions for safe innovation*."[2]

On 11 July 2022, the FSB issued a public communication[3] that highlights the potential risks and threats arising from crypto-assets; stresses that crypto-asset activities do not operate in a regulation-free space; expresses concern about lack of conformance with existing standards, applicable rules and regulations; and states that crypto-assets providers must not commence operations in any jurisdiction unless any such service provider meets all applicable regulatory requirements. The communication also reaffirms the FSB's role in facilitating cooperation among jurisdictional financial authorities and international standard-setting bodies (SSBs) to ensure that crypto-asset activities and markets are subject to effective regulation and oversight commensurate with the risks they may pose, while supporting responsible innovation and providing sufficient flexibility for jurisdictions to implement domestic approaches.

In October 2022, the FSB published a consultative report on regulation, supervision and oversight of crypto-asset activities and markets, including a set of high-level recommendations. This final report takes into account feedback from the public consultation report and stakeholder outreach. In light of the events that took place in crypto-asset markets in 2022 and early 2023 and the potential threat to the wider financial system, the report also reflects enhancements of key areas. This final report contains solely the high-level recommendations and the glossary, which are core components of the FSB framework. The analysis presented in Sections 2 and 3 of the consultative report remains valid, and thus is not repeated in this final report.

Whereas the FSB's review of its High-level Recommendations on the Regulation, Supervision and Oversight of "Global Stablecoin" Arrangements that is published alongside this report is focused on stablecoins as a subset of crypto-assets, this report's focus is on the crypto-asset activities and markets more broadly.

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1 FSB (2020): *Regulation, Supervision and Oversight of "Global Stablecoin" Arrangements: Final Report and High-Level Recommendations*, October.

2 *Communiqué G20 Finance Ministers and Central Bank Governors Meeting*, 17-18 February, 2022. Jakarta, Indonesia.

3 FSB (2022): *FSB Statement on International Regulation and Supervision of Crypto-asset Activities*, July.

In line with the mandate of the FSB, the focus of this report is on regulatory, supervisory and oversight issues relating to crypto-assets to help foster safe innovation. The report therefore does not comprehensively address all specific risk categories related to crypto-asset activities, such as: Anti-Money Laundering/Combating the Financing of Terrorism (AML/CFT); data privacy; cyber security; consumer and investor protection; market integrity; competition policy; taxation; monetary policy; monetary sovereignty; and other macroeconomic concerns.

The FSB has been working closely with the International Monetary Fund (IMF), World Bank, the Organization for Economic Cooperation and Development (OECD), the Basel Committee on Banking Supervision (BCBS), the Bank for International Settlements' Committee on Payments and Market Infrastructures (CPMI), the International Organization of Securities Commissions (IOSCO), and the Financial Action Task force (FATF) to ensure that the work underway regarding the monitoring and regulation of crypto-asset activities and markets is coordinated and mutually supportive.

2. Objectives, scope and follow-up

2.1. Objectives and scope

The proposed recommendations seek to promote the comprehensiveness and greater international consistency of regulatory and supervisory approaches to crypto-asset activities and markets, including crypto-asset issuers and service providers[4]. The recommendations are intended to be flexible so that they can be incorporated into the wide variety of regulatory frameworks potentially applicable to crypto-asset activities and markets around the world, and do not displace existing applicable regulatory, supervisory and oversight frameworks. These recommendations apply to any type of crypto-assets in any jurisdiction and should inform the regulation of any type of crypto-asset activities, including those conducted through so-called decentralized finance (DeFi) protocols, that pose, or potentially pose, risks to financial stability, both individually and collectively. Central bank digital currencies (CBDCs) are not subject to these recommendations. These recommendations should be applied to crypto-asset issuers and service providers in a way that is proportionate to their risk, size, complexity and systemic importance.

Crypto-asset activities that meet the definition of a Global Stablecoin ("GSC") arrangement, as defined in the FSB High-level Recommendations for the Regulation, Supervision and Oversight of "Global Stablecoin" Arrangements[5] should also be subject to regulatory and supervisory approaches that implement the FSB's recommendations for GSC arrangements. Authorities may choose to apply relevant High-level Recommendations on GSC arrangements as appropriate to stablecoin arrangements more widely, taking into account the risk, size and complexity of those stablecoins.

Crypto-asset markets are fast evolving and could reach a point where they represent a threat to global financial stability due to their scale, structural vulnerabilities and increasing

4 Crypto-asset issuers and service providers are defined in the Annex of this document.

5 FSB (2020). An updated version of the recommendations is being consulted on in parallel with this report.

interconnectedness with the traditional financial system. The rapid evolution and international nature of these markets also raise the potential for regulatory gaps, fragmentation and arbitrage. Although the extent and nature of use of crypto-assets varies somewhat across jurisdictions, financial stability risks could rapidly escalate, underscoring the need for both timely and pre-emptive evaluation of possible policy responses, as well as regulatory action where existing requirements apply. Authorities need to be ready to regulate, supervise, and oversee these activities and the associated issuers and service providers that have the potential to pose risks to financial stability.

The recommendations are addressed to financial regulatory, supervisory and oversight authorities at a jurisdictional level. They set out the key objectives that an effective regulatory and supervisory framework should achieve but are high-level and flexible so that they can be incorporated into a wide variety of regulatory frameworks. The recommendations establish a global regulatory baseline, and some jurisdictions may also decide to take more restrictive regulatory measures.[6] Their aim is to promote a regulatory, supervisory and oversight framework that is technology-neutral and focuses on underlying activities and risks.

The recommendations focus on addressing risks to financial stability, and they do not comprehensively cover all specific risk categories related to crypto-asset activities, such as: AML/CFT; data privacy; cyber security; consumer and investor protection; market integrity; competition policy; taxation; monetary policy; monetary sovereignty and other macroeconomic concerns. A comprehensive supervisory and regulatory framework for crypto-asset activities that effectively addresses these other important policy objectives will improve the stability of the crypto-asset market and thereby reduce the risks of negative spillovers to the wider financial system. The FSB therefore supports related efforts by SSBs and authorities to ensure such a comprehensive regulatory framework for the crypto-asset ecosystem. For example, regulations that address investor protection and market integrity can also reduce financial stability risk by increasing regulatory and public transparency.

Authorities should seek to apply the recommendations consistent with their respective mandates. An effective application of these recommendations by relevant authorities in jurisdictions in which the crypto-asset activities, issuers and service providers are active will help to ensure a comprehensive regulatory coverage and reduce the scope for regulatory arbitrage or evasion.

2.2. Follow-up and review

The FSB and the SSBs will continue to encourage consistency and a common understanding of the key elements of comprehensive regulatory, supervisory and oversight frameworks for crypto-asset activities and markets, and will support authorities in implementing the recommendations as crypto-asset activities and markets evolve.

The FSB will, in close cooperation with relevant SSBs, take the appropriate actions to (i) continue to coordinate international regulatory, supervisory and oversight approaches for crypto-asset activities to ensure they are comprehensive, consistent and complementary, including by

6 For example, jurisdictions could choose to prohibit certain or all crypto-asset activities.

considering the findings of the vulnerability analysis work on DeFi, crypto-asset service providers that combine multiple functions and whether additional policy work is warranted; and (ii) conduct a review of the implementation of the recommendations by end-2025 that may help determine whether a further review of the recommendations [or further action to promote implementation] may be necessary.

Table 1 shows the indicative timelines for this work following the publication of the high-level recommendations.

Table 1: Follow-up work to the FSB consultative report and recommendations

Continue to coordinate international regulatory and supervisory approaches for crypto-asset activities	
The FSB will continue to coordinate international regulatory and supervisory approaches for crypto-asset activities to ensure that they are comprehensive, consistent and complementary. Depending on the outcome of the FSB's analysis of potential risks to financial stability stemming from DeFi and crypto-asset service providers that combine multiple functions, the FSB will consider the regulatory implications in these areas and assess whether additional policy work is warranted.	• By end-2024
Review the implementation of the recommendations	
FSB will, in consultation with relevant SSBs and international organisations, conduct a review of the implementation of recommendations in FSB jurisdictions and assess the need to update the recommendations.	• By end-2025

3. Final recommendations

Recommendation 1: Regulatory powers and tools

Authorities should have and utilise the appropriate powers and tools, and adequate resources to regulate, supervise, and oversee crypto-asset activities and markets, and enforce relevant laws and regulations effectively, as appropriate.

Authorities within a jurisdiction, either independently or collectively, should have and utilise the appropriate powers and tools and adequate resources to regulate, supervise, and oversee crypto-asset activities and markets as appropriate.

Authorities should require that crypto-asset issuers and service providers meet all applicable regulatory, supervisory and oversight requirements of a particular jurisdiction before commencing any operations in that jurisdiction and adapt to new regulatory requirements as necessary and appropriate.

Authorities should have the powers and capabilities to enforce applicable regulatory, supervisory and oversight requirements, including authorisation and licensing requirements, and the ability to undertake inspections or examinations.

When crypto-asset issuers or service providers are not complying with applicable laws or regulations, authorities should have the powers and capabilities to require corrective actions and

take enforcement actions as appropriate. Authorities should ensure they have the powers and capabilities to address the risks arising from efforts by crypto-asset issuers and service providers to evade regulation and oversight by operating in foreign jurisdictions. This may include consideration of restrictions on the access by domestic users to foreign crypto-asset service providers and markets, or the provision of services by foreign crypto-asset issuers and service providers, when they do not comply with applicable domestic regulations and are not regulated to international standards.

Authorities should require crypto-asset service providers to have a well-founded, clear, transparent and enforceable legal basis for each material aspect of their activities in all relevant jurisdictions.

Recommendation 2: General regulatory framework

Authorities should apply comprehensive and effective regulation, supervision, and oversight to crypto-asset activities and markets – including crypto-asset issuers and service providers – on a functional basis and proportionate to the financial stability risk they pose, or potentially pose, and consistent with authorities' respective mandates in line with the principle "same activity, same risk, same regulation".

Authorities should have in place comprehensive regulatory rules and policies applicable to crypto-asset activities, issuers and service providers proportionate to their risk, size, complexity and systemic importance, and consistent with the economic functions they perform in line with the principle of "same activity, same risk, same regulation" and relevant international standards while also taking into account the specific risks associated with crypto-asset activities. Given the fast-evolving nature of crypto-asset activities and markets and the potential for financial stability risks to rapidly emerge or escalate, authorities should be ready to regulate and supervise crypto-asset activities and markets, that have the potential to pose risks to financial stability.

Consistent with past approaches to technological change, authorities should assess whether existing regulatory, supervisory and oversight requirements adequately address the financial stability risks of crypto-asset activities, including any emerging or new risks that may arise and, if needed, clarify or supplement existing regulatory, supervisory and oversight requirements. In cases when crypto-asset activities outside the scope of financial regulation may pose risks to financial stability, authorities should, as needed, seek to expand or adjust their regulatory perimeter, as appropriate.

The assessment of potential financial stability risks should take into account the interconnectedness between the crypto-asset market and the wider financial system, the overall size and nature of the activities being conducted (including the degree of financial intermediation, leverage, credit, liquidity and maturity transformation), as well as of the risk of spillovers into other jurisdictions.

Authorities should target regulatory outcomes in the crypto-asset market equivalent to those in the traditional financial market so as not to incentivise the circumvention of regulation through

the migration of traditional financial activities to crypto-asset markets. To this end, authorities should consider relevant sectoral standards and policies[7].

Regardless of whether crypto-asset activities are conducted in purportedly decentralised structures or other ways that frustrate the identification of a responsible entity or an issuer of the crypto-assets, authorities should adopt or have in place a regulatory approach that aims at adequate protection for all relevant parties, including consumers and investors, and at achieving the same regulatory outcome.

Recommendation 3: Cross-border cooperation, coordination and information sharing

Authorities should cooperate and coordinate with each other, both domestically and internationally, to foster efficient and effective communication, information sharing and consultation in order to support each other as appropriate in fulfilling their respective mandates and to encourage consistency of regulatory and supervisory outcomes.

Authorities should cooperate in the regulation, supervision and oversight of crypto-asset activities and markets. Authorities should use existing cooperation and information sharing arrangements, such as memoranda of understanding, or ad-hoc arrangements, to the extent practicable, or consider establishing new arrangements that may encompass additional subject areas or jurisdictional authorities and that consider the cross-sectoral nature of some activities.

Cross-border cooperation and information sharing among authorities should aim to facilitate a shared understanding of the risks and activities of crypto-assets, issuers and service providers across jurisdictions in normal times and in times of stress. Authorities should have in place adequate information sharing about crypto-asset service providers and issuers which are part of a group operating in multiple jurisdictions. Authorities should share information and cooperate in a timely and effective way with respect to crypto-asset issuers and service providers that are in financial or operational distress and which may affect a wider range of jurisdictions than where they primarily operate.

Authorities should take additional steps to collaborate with authorities in relevant jurisdictions when they host crypto-asset issuers and service providers with a global reach, taking into account the risk of spillovers into other jurisdictions.

To foster effective cross-border cooperation and coordination, the FSB and the SSBs will continue to promote consistency and a common understanding of key elements of regulatory, supervisory and oversight frameworks for crypto-asset activities and markets. The FSB will continue to monitor issues related to the cross-border nature of crypto-assets and consider mechanisms that facilitate exchange of information about the level of compliance by service providers that are operating across borders and out of jurisdictions that have not implemented a regulatory framework for crypto-asset activities consistent with the FSB high-level recommendations and international standards.

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[7] For instance, the IOSCO Objectives and Principles of Securities Regulation, CPMI-IOSCO Principles for financial market infrastructures, the Basel Framework, and FATF standards, in particular FATF Recommendations 15 and 16.

Recommendation 4: Governance

Authorities, as appropriate, should require that crypto-asset issuers and service providers have in place and disclose a comprehensive governance framework with clear and direct lines of responsibility and accountability for all functions and activities they are conducting. The governance framework should be proportionate to their risk, size, complexity and systemic importance, and to the financial stability risk that may be posed by activity or market in which the crypto-asset issuers and service providers are participating. It should provide for clear and direct lines of responsibility and accountability for the functions and activities they are conducting.

Authorities should require crypto-asset issuers and service providers to have a robust governance framework. The framework should be proportionate to their risk, size, complexity and systemic importance, and to the financial stability risk that may be posed by the activity or market in which they are participating. It should include clear and direct lines of responsibility and accountability, clear definition of the roles and responsibilities of the management body and the decision-making process, including procedures for identifying, addressing and managing conflicts of interest.

Where crypto-asset activities are conducted in ways that may frustrate the identification of the responsible entity or affiliated entities, such as through so-called DeFi protocols or setting up complex corporate structures, such conduct of activities must not undermine robust governance and accountability arrangements. Authorities should require compliance with rules and regulations for effective governance irrespective of the structures of activities and technology used to conduct the crypto-asset activities.

Recommendation 5: Risk management

Authorities, as appropriate, should require crypto-asset service providers to have an effective risk management framework in place that comprehensively addresses all material risks associated with their activities. The framework should be proportionate to the risk, size, complexity, and systemic importance, and to the financial stability risk that may be posed by the activity or market in which they are participating. Authorities should, to the extent necessary to achieve regulatory outcomes comparable to those in traditional finance, require crypto-asset issuers to address the financial stability risk that may be posed by the activity or market in which they are participating.

Authorities should understand the different risk profiles of crypto-asset issuers and service providers and require them, as appropriate, to establish a risk management framework that is proportionate to their risk, size, complexity, and systemic importance, and to the financial stability risk that may be posed by the activity or market in which they are participating.

Authorities should expect crypto-asset issuers and service providers to be directed by a management which is qualified and of good repute (for example, by applying “fit and proper” standards, where applicable); allocate adequate resources to risk management and other control functions (i.e. compliance and internal audit); and ensure that these control functions are carried out independently of business activities. Authorities should pay particular attention to the compliance functions of each service provider or issuer, which should be expected to monitor all

of the activities of that service provider or issuer, to identify and address any non-compliance issues and deficiencies, and to ensure adherence to applicable laws and regulations.

Authorities should expect crypto-asset issuers and service providers to act honestly and fairly and require them to communicate with users and relevant stakeholders in a clear and not misleading manner, and identify, disclose, manage and, to the extent possible, prevent any conflict of interests.

Authorities, as appropriate, should require crypto-asset issuers and crypto-asset service providers, proportionate to their risk, size, complexity, systemic importance, and to the financial stability risk that may be posed by the activity or market in which they are participating, to identify, measure, evaluate, monitor, report, and control all material risks. Authorities should require crypto-asset service providers to effectively identify and manage risks arising from leverage and credit, liquidity, operational, compliance, and maturity transformation. Authorities should require crypto-asset service providers to have rules and procedures to effectively monitor the transactions they conduct. Authorities should also have in place rules, policies and enforcement tools that comprehensively address these risks both in normal times and in times of stress.

Authorities should, as appropriate, consider applying both prudential and market conduct regulatory tools. Authorities should pay particular attention to technological risks associated with crypto-asset activities. Authorities should require that crypto-asset service providers that are part of an internationally active group properly identify, monitor and manage risks arising from exposures to other parts of the group.

Authorities, as appropriate, should require crypto-asset issuers and crypto-asset service providers, proportionate to their risk, size, complexity, systemic importance, and to the financial stability risk that may be posed by the activity or market in which they are participating, to establish effective contingency arrangements (including robust and credible recovery plans where warranted) and business continuity planning.

Authorities should ensure that crypto-asset issuers and crypto-asset service providers put appropriate AML/CFT measures in place consistent with FATF Standards, including requirements to comply with the FATF ‘”travel rule”.

Taking into account the jurisdictional differences in legal frameworks regarding ownership rights to crypto-assets held by a crypto-asset service provider, authorities should require all crypto-asset service providers involved in holding or safeguarding crypto-assets to have in place adequate controls, to disclose to customers the rights of ownership they have, and to protect customers’ ownership rights, including through segregation and record-keeping requirements. These arrangements should minimise the risk of loss, misuse of or delayed access to assets, including in the event of a service provider’s insolvency. The same requirements should apply when the activities involved in holding or safeguarding crypto-assets are outsourced to a third party.

Authorities should require crypto-asset service providers facilitating trading to ensure that their operations are resilient and transparent and should implement and maintain clear and transparent operating rules for the trading platform.

Recommendation 6: Data collection, recording and reporting

Authorities, as appropriate, should require that crypto-asset issuers and service providers have in place robust frameworks, including systems and processes, for collecting, storing, safeguarding, and the timely and accurate reporting of data, including relevant policies, procedures and infrastructures needed, in each case proportionate to their risk, size, complexity and systemic importance. Authorities should have access to the data as necessary and appropriate to fulfil their regulatory, supervisory and oversight mandates.

Authorities should require that crypto-asset issuers and service providers, proportionate to their risk, size, complexity and systemic importance, have data management systems that record and safeguard relevant data and information collected and produced in the course of their operations, with adequate controls in place to safeguard the integrity and security of relevant data and conform to applicable regulation, including on data retention, data security and data privacy. Appropriate infrastructures should be maintained in order to ensure data quality and reliability and have in place well-defined procedures to monitor data quality and rectify poor data. Authorities should require crypto-asset service providers to have measures in pace to ensure the completeness, accuracy and reliability of data.

Authorities should have full, timely, complete, and ongoing access to relevant data and information, wherever the data is located, to enable them to regulate, supervise and oversee the functions and activities of the crypto-asset activities and markets, considering the level and nature of the risks posed. Authorities should seek to address any impediments to relevant data access or limitations of the data.

Authorities may leverage existing efforts to promote consistent and comparable data collection and reporting based on activity types and economic functions or consider developing new reporting frameworks or policies to support data collection and sharing, as appropriate, across relevant authorities and jurisdictions.

Authorities should seek to promote the public understanding of crypto-asset markets. For service providers that facilitate a wide range of trading services and a large size of trading volume, authorities should assess their ability to access data regarding, but not limited to, the instruments most frequently traded, the principal amounts traded, and the largest counterparties and intermediaries, and the extent to which these data should be made more widely available to the public or publicly disseminated.

Recommendation 7: Disclosures

Authorities should require that crypto-asset issuers and service providers disclose to users and relevant stakeholders comprehensive, clear and transparent information regarding their governance framework, operations, risk profiles and financial conditions, as well as the products they provide and activities they conduct.

Authorities should require that crypto-asset issuers and service providers make available to users and relevant stakeholders, including customers, investors or shareholders, all necessary information regarding how they operate, how they transact, the risk features of their products,

and how they manage and mitigate any potential risks in an understandable manner for the intended audiences. This should include, as appropriate, the governance structure and procedures related to the main activities offered[8] and important conflict of interests emanating from crypto-asset activities.

Authorities should require that crypto-asset issuers and service providers adequately disclose the information related to the product structure and the operation of the activities they conduct. This may include, for example, a prospectus or an equivalent document from a crypto-asset issuer.

Authorities should require the service provider to provide full and accurate disclosure to any client for whom it is providing custody services of the terms and conditions of the custodial relationship and the risks that could be faced by the client if the custodian were to enter bankruptcy. This disclosure should include the ownership rights retained by the client, the safeguards implemented by the service provider for client assets, the existence of any outsourcing arrangements, and any circumstances where a crypto-asset issuer or service provider might not provide timely redemption or withdrawal of assets. Where ownership rights are intended to be transferred to the crypto-asset service provider, authorities should require the service provider to clearly disclose the arrangement to the customer and to obtain the customer's explicit prior consent to the transfer.

Authorities should require crypto-asset issuers and service providers to disclose any material risks associated with the underlying technologies, such as cyber security risk, as well as environmental and climate risks and impacts, as appropriate and in line with jurisdictional legal frameworks.

Recommendation 8: Addressing financial stability risks arising from interconnections and interdependencies

Authorities should identify and monitor the relevant interconnections, both within the crypto-asset ecosystem, as well as between the crypto-asset ecosystem and the wider financial system. Authorities should address financial stability risks that arise from these interconnections and interdependencies.

Authorities should identify and address potential financial stability risks that may originate from or be transmitted or amplified by the crypto-asset ecosystem. Authorities should seek to identify and monitor on an ongoing basis interlinkages and interdependencies among different parts of the crypto-asset ecosystem and assess the aggregated risk arising from interlinkages between the crypto-asset ecosystem, the wider financial system and the real economy.

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8 For example, key decision-making procedures and voting mechanisms, clear and accurate description of responsibilities and rights of all stakeholders, important change of protocols, available dispute mechanisms or procedures for seeking redress or lodging complaints, composition of balance sheet items, financial conditions, regulatory incidents and penalties Where relevant, this information should also include redemption rights and composition of reserve assets for those crypto-assets that aim to maintain a stable value relative to a specified asset, or a pool or basket of assets.

As a component of monitoring interlinkages between the crypto-asset ecosystem and the wider financial system, authorities should consider the scale of crypto-asset activities and whether this presents systemic risk to the wider financial system.

Where financial stability risks arise from traditional financial institutions' exposures to crypto-assets, authorities should address these risks in line with the recommendations and based on frameworks developed by the SSBs for these institutions.

Recommendation 9: Comprehensive regulation of crypto-asset service providers with multiple functions

Authorities should ensure that crypto-asset service providers and their affiliates that combine multiple functions and activities, where permissible, are subject to appropriate regulation, supervision and oversight that comprehensively address the risks associated with individual functions and the risks arising from the combination of functions, including but not limited to requirements regarding conflicts of interest and separation of certain functions, activities, or incorporation, as appropriate.

Certain crypto-asset service providers, including groups of affiliated service providers, may be undertaking a variety of functions, including facilitating transactions, settlement and clearing, non-custodial and custodial wallet provisioning (including the sale of software and hardware for non-custodial wallets), market-making, offering investment vehicles, lending and borrowing, and proprietary trading and issuance. In some jurisdictions, certain combinations are not permitted. Where combinations are permitted, relevant authorities should work to ensure that these service providers, including groups of affiliated service providers, are subject to robust and comprehensive regulation, supervision and oversight that address the risks arising from the combination of multiple activities and functions, in particular those that fall under different sectoral regimes, with strong protection for investors and consumers. Authorities should consider requirements that address not only risks on a standalone basis, but also additional risks and additional conflicts of interest when those functions and activities are conducted concurrently.

Authorities should require that crypto-asset service providers, including groups of affiliated service providers that, where permissible, combine multiple functions, have transparent organisational and managerial structures, which are consistent with their overall strategy and risk profile and which are well understood by the board and senior management of the service provider, affiliated service providers, regulators and customers.

Authorities should consider whether and, if so, how combinations of multiple functions can be appropriately regulated within a single entity or group of affiliated entities. To the extent that such combinations are a result of non-compliance with existing regulations or will generate acute conflicts of interest which, as determined by authorities, cannot be effectively managed, authorities should apply robust measures as appropriate and in line with jurisdictional legal frameworks, including legal disaggregation and separation of certain functions. Authorities should consider relevant sectoral standards developed by SSBs when they require segregation of functions. Where multiple functions are permitted to exist within a crypto-asset service provider or group of affiliated service providers, authorities should require that such service providers have adequate policies and processes to identify, mitigate and manage actual or

potential conflicts of interests, including prevention of any abuse resulting from concentrated control, management of related-party transactions, in particular those involving affiliated companies, and transparency of related-party transactions. Authorities should consider additional prudential requirements if appropriate to address additional risks or conflicts of interest from this setup.

Authorities should pay particular attention to service providers combining multiple functions that engage in facilitating custody, trading, settlement, lending, and borrowing or proprietary trading, and should apply regulatory measures that are designed for the adequate segregation of risks. For example, this may include legal disaggregation and separation of certain functions.

Cross-border and cross-sectoral information sharing about service providers combining multiple functions and operating across borders and sectors is particularly important. Authorities should share information, as appropriate, to minimize financial distress of these intermediaries from spilling over to other jurisdictions or sectors of the financial system.

Glossary

Blockchain

A form of distributed ledger in which details of transactions are held in the ledger in the form of blocks of information. A block of new information is appended to the chain of pre-existing blocks via a computerised process by which transactions are validated.

Crypto-asset

A digital asset (issued by the private sector) that depends primarily on cryptography and distributed ledger or similar technology.

Crypto-asset ecosystem

The entire ecosystem that encompasses all crypto-asset activities, market and participants.

Crypto-asset issuer

An entity, person, or other structure that creates new crypto-assets.

Crypto-asset market

Any place or system that provides buyers and sellers the means to trade crypto-assets and the associated instruments, including lending, structured investment products, and derivatives. Crypto-asset markets facilitate the interaction between those who wish to offer and sell and those who wish to invest.

Crypto-asset services

Services relating to crypto-assets that may include, but are not limited to, distribution, placement, facilitating exchange between crypto-assets or against fiat currencies, custody, provisioning of non-custodial wallets, facilitating crypto-asset trading, borrowing or lending, and acting as a broker-dealer or investment adviser.

Crypto-asset service providers

Individuals and entities that provide crypto-asset services.

Crypto-asset activities

Activities serviced by a crypto-asset issuer or crypto-asset service provider.

Crypto-asset trading platform:

Any platform where crypto-assets can be bought and sold, regardless of the platform's legal status.

Decentralised Finance (DeFi)

A set of alternative financial markets, products and systems that operate using crypto-assets and ‘smart contracts’ (software) built using distributed ledger or similar technology.

DeFi protocols

A specialised autonomous system of rules that creates a program designed to perform financial functions.

Digital asset

A digital representation of value or contractual rights which can be used for payment or investment purposes.

Global stablecoin (GSC)

A stablecoin with an existing or potential reach and use across multiple jurisdictions and which could become systemically important in and across one or many jurisdictions, including as a means of making payments and/or store of value.

Stablecoin

A crypto-asset that aims to maintain a stable value relative to a specified asset, or a pool or basket of assets.

Wallet

An application or device for storing the cryptographic keys providing access to crypto-assets. A hot wallet is connected to the internet and usually takes the form of software for the user, while a cold wallet is a hardware that is not connected to the internet and stores the cryptographic keys.

Custodial wallet

A service in which crypto-assets are held by a service provider. A user interacts with the service provider to manage the user’s crypto-assets. A custodial wallet is also known as a “hosted wallet”.

Non-custodial wallet

Software or hardware that stores cryptographic keys for a user, making the user’s crypto-assets accessible only to the user, and allowing the user to interact directly with the blockchain and the blockchain-based finance applications. A non-custodial wallet is also known as an “unhosted wallet”.

FSB FINANCIAL STABILITY BOARD

High-level Recommendations for the Regulation, Supervision and Oversight of Global Stablecoin Arrangements

Final report



17 July 2023

The Financial Stability Board (FSB) coordinates at the international level the work of national financial authorities and international standard-setting bodies in order to develop and promote the implementation of effective regulatory, supervisory and other financial sector policies. Its mandate is set out in the FSB Charter, which governs the policymaking and related activities of the FSB. These activities, including any decisions reached in their context, shall not be binding or give rise to any legal rights or obligations.

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1. Introduction

In October 2020, the FSB issued a report with a set of 10 high-level recommendations on the regulation, supervision and oversight of so-called "global stablecoin" (GSC) arrangements ("High-level Recommendations"). The G20 endorsed the High-level Recommendations, and in October 2021 the FSB provided a status update on the progress made on their implementation.

In October 2022, the FSB published a review of its High-level Recommendations, including how any gaps identified could be addressed by existing frameworks, considering recent market and policy developments.[1] This final report takes into account feedback from the public consultation report and stakeholder outreach.

The High-level Recommendations seek to promote consistent and effective regulation, supervision and oversight of GSCs across jurisdictions to address the potential financial stability risks posed by GSCs, both at the domestic and international level, while supporting responsible innovation and providing sufficient flexibility for jurisdictions to implement domestic approaches. The recommendations take a broad approach to global stablecoins and are intended to be flexible so that they can be incorporated into the wide variety of regulatory frameworks potentially applicable to GSCs around the world. International standards designed for specific sectors focus on distinct functions within the remit of the relevant standard setting bodies (SSBs). Where international sectoral standards apply to a GSC for a particular economic function, those standards will address risks specific to the economic function and, as such, authorities should implement those international standards.

There is no universally agreed legal or regulatory definition of stablecoin. The term stablecoin used in this report does not denote a distinct legal or regulatory classification. Importantly, the use of the term "stablecoin" in this report is not intended to affirm or imply that its value is stable. Rather, the term is used because it is commonly employed by market participants and authorities. The FSB's 2020 report, "Regulation, Supervision and Oversight of 'Global Stablecoin' Arrangements" described three characteristics that distinguish a GSC from other crypto-assets and other stablecoins. Those characteristics include: (i) the existence of a stabilisation mechanism[2], (ii) the usability as a means of payment and/or store of value[3], and (iii) the potential reach and adoption across multiple jurisdictions. The first two characteristics (the existence of a stabilisation mechanism and usability as a means of payment and/or store of value), and the unique risks that these characteristics pose, distinguish stablecoins from other crypto-assets. The third, the potential reach and adoption across multiple jurisdictions, differentiates GSCs from other stablecoins.

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1 For more detail on the findings from this review, including the risks posed by GSC arrangements, and how any regulatory gaps identified could be addressed by the revised recommendations and existing frameworks, see Review of the FSB High-level Recommendations of the Regulation, Supervision and Oversight of "Global Stablecoin" Arrangement – Consultative Report (2022)

2 Recommendation 9 requires GSCs to have an effective stabilisation mechanism. Therefore, so-called algorithmic stablecoins do not meet the High-Level Recommendations for GSCs in particular as they do not use an effective stabilisation method.

3 To be useable as a means of payment and/or store of value, a stablecoin arrangement typically provides three core functions. These core functions include (i) issuance, redemption and stabilisation of the value of the coins; (ii) transfer of coins; and (iii) interaction with coin users for storing and exchanging coins.

2. Objectives and scope and follow up

These High-level Recommendations seek to promote consistent and effective regulation, supervision and oversight of GSCs and stablecoins with the potential to become GSCs across jurisdictions (hereafter, unless otherwise specified, they are collectively referred to as "GSCs") to address the financial stability risks posed, both at the domestic and international level, while supporting responsible innovation and providing sufficient flexibility for jurisdictions to implement domestic approaches. Central bank digital currencies (CBDCs) are not subject to these recommendations.

The recommendations are intended to be flexible so that they can be incorporated into the wide variety of regulatory frameworks potentially applicable to GSCs around the world, and do not displace existing applicable regulatory, supervisory and oversight frameworks. They aim to promote a regulatory, supervisory and oversight environment that is technology neutral and focuses on underlying activities and risks.

The recommendations are addressed to financial regulatory, supervisory and oversight authorities at a jurisdictional level. They should be applied by individual authorities to the extent they fall within the authorities' remits.

Grounded in an assessment of a GSC arrangement's economic functions and the principle of "same activity, same risk, same regulation", and focused on regulatory objectives and outcomes, authorities should apply and, if necessary, develop effective regulatory, supervisory and oversight approaches and cross-border cooperation mechanisms within their respective mandate and legal frameworks.

At the same time, the recommendations set out expectations for providers of services and activities within the GSC arrangements and can serve as a basis for authorities' active engagement with stakeholders on GSC-related risks and how these are addressed.

2.1. Scope

The recommendations focus on addressing risks to financial stability and therefore do not comprehensively cover important issues such as anti-money laundering and counter-terrorist financing (AML/CFT), data privacy, cyber security consumer and investor protection, market integrity, competition policy, taxation, monetary policy, monetary sovereignty, currency substitution or other macroeconomic concerns. However, a comprehensive supervisory and regulatory framework for GSC arrangements should effectively address these other issues in addition to financial stability risks.

The recommendations apply to any GSC in any jurisdiction and help authorities to address activities and services within GSC arrangements that may fall outside some jurisdictions' traditional regulatory perimeter. Section 4.4 includes potential elements that could be used to determine whether a stablecoin qualifies as a GSC. Consistent application of these recommendations by all relevant authorities in jurisdictions in which GSC arrangements are active may help ensure comprehensive regulatory coverage and reduce the scope for regulatory arbitrage.

The recommendations establish a global regulatory baseline and some jurisdictions may also decide to take more restrictive regulatory measures.[4] In addition, while focusing on GSCs identified based on this report and the criteria set out in Section 4.4, authorities may choose to apply relevant High-level Recommendations as appropriate to stablecoin arrangements more widely, taking into account the size, complexity and risks of those stablecoins.

The High-level Recommendations complement and are intended to inform any potential updates to international sectoral standards and principles. In particular, they complement the FSB's recommendations for crypto-assets and markets that should apply to any crypto-asset activity that poses financial stability risks, including stablecoins. Authorities should rely on sectoral standards and principles for cross-border cooperation where GSC arrangements perform the same economic function as existing regulated activities covered by those standards. These include, for example, the IOSCO Principles regarding Cross-Border Supervisory Cooperation, the CPMI-IOSCO Principles for Financial Market Infrastructures, including the Responsibilities of Authorities and particularly Responsibility E, the March 2020 IOSCO Report on Global Stablecoin Initiatives, the Financial Action Task Force (FATF) standards, in particular Recommendation 15, and the relevant principles applicable to cross-border banking supervision and crisis management of the Basel Committee on Banking Supervision (BCBS) and the FSB. Efforts by the SSBs to review, and where appropriate adjust, their standards to take into account the novel features of stablecoins can further promote international consistency and reduce the risk of regulatory arbitrage or regulatory underlaps.

2.2. Follow-up and review

The FSB, in close cooperation with relevant SSBs, will take the appropriate actions to (i) continue to coordinate international regulatory, supervisory and oversight approaches for global stablecoin arrangements to ensure they are comprehensive, consistent and complementary; and (ii) conduct a review of the implementation of the recommendations by end-2025 that may help determine whether a further review of the recommendations or further action to promote implementation may be necessary.

3. Final recommendations

Recommendation 1: Authorities' readiness to regulate and supervise global stablecoin arrangements

Authorities should have and utilise the appropriate powers and tools, and adequate resources, to comprehensively regulate, supervise, and oversee a GSC arrangement and its associated functions and activities, and enforce relevant laws and regulations effectively.

Authorities within a jurisdiction, either independently or collectively, should have and utilise the powers and capabilities to, as applicable, regulate, supervise, oversee and, if necessary or

[4] For example, jurisdictions could choose to prohibit some or all stablecoin activities.

appropriate, effectively prohibit stablecoin activities being conducted and stablecoin services being offered to users in or from their jurisdiction, consistent with jurisdictions' laws and regulations. This may include, for example, activities and services related to the governance and control of the stablecoin arrangement, operating its infrastructure, issuing and redeeming stablecoins, managing stablecoin reserve assets, providing custody or trust services for those assets, trading and exchanging stablecoins, or storing the keys providing access to stablecoins. Application of an authority's powers to regulate, supervise, and oversee GSC arrangements should be commensurate with their existing or potential size, complexity, risk and/or extent of use as a means of payment and/or store of value.

Authorities should ensure that they are ready to use their powers and capabilities to regulate, supervise and oversee GSC arrangements. Authorities should consider the potential for stablecoins to rapidly scale and become a GSC and should ensure appropriate monitoring of GSC activities and any significant changes to the way those activities are conducted. Authorities should have timely access to relevant information sufficient to conduct effective regulation, supervision and oversight.

Authorities' powers should extend to entities and persons that are engaged in activities of GSCs in their jurisdictions and fall within the scope of their authority and mandate. Authorities should evaluate, identify, and clarify which authorities have responsibility for each activity of a GSC arrangement, as appropriate.

Authorities should identify and address any significant gaps in their regulatory, supervisory and oversight frameworks through changes in regulations, or policy, as appropriate. In some jurisdictions, legislative changes may be necessary or appropriate to address those gaps.

Authorities should have the powers and capabilities to enforce applicable regulatory, supervisory and oversight requirements, including the ability to undertake inspections or examinations, and, when necessary or appropriate, require corrective actions and take enforcement measures. To do so, authorities should be provided with or obtain sufficient information regarding the technology and legal obligations underpinning the GSC arrangements.

Authorities should be able to identify the legal entities or persons responsible for the relevant activities and to assess the ability of the GSC arrangement to implement corrective actions.

Authorities should have the ability to mitigate risks, or prohibit the use of certain or specific stablecoins in their jurisdictions, consistent with jurisdictions' laws and regulations, where these do not meet the applicable regulatory, supervisory, and oversight requirements.

Recommendation 2: Comprehensive oversight of GSC activities and functions

Authorities should apply comprehensive and effective regulatory, supervisory and oversight requirements consistent with international standards to GSC arrangements on a functional basis and proportionate to their risks insofar as such requirements are consistent with their respective mandates.

To promote a technology neutral approach that enables comprehensive oversight of GSC's functions and activities and mitigates regulatory arbitrage, authorities should focus on the functions performed by the GSC arrangement and risks posed and apply the appropriate

regulatory framework, consistent with international standards, in the same manner as they would apply it to entities and persons performing the same functions or activities, and posing the same risks ("same activity, same risk, same regulation").

This includes the relevant regulation, standards and rules for e-money issuers, remittance companies, financial market infrastructures including payment systems, collective investment schemes, and deposit-taking and/or securities issuance and trading activities. This also includes market integrity, consumer and investor protection arrangements, appropriate safeguards, such as pre- and post-trade transparency obligations, rules on conflicts of interest (including for different service providers such as, the reserve asset custodian, stablecoin trading, lending and borrowing platforms), disclosure requirements, robust systems and controls for platforms where the GSC is traded, rules that allocate responsibility in the event of unauthorised transactions and fraud, and rules governing the irrevocability of a transfer orders ("settlement finality").

Trading platforms and other intermediaries and service providers

Where a GSC arrangement relies on trading platforms or other intermediaries to perform critical functions, including some or all of its stabilisation function, authorities should require that those intermediaries fall within the regulatory, supervisory and oversight perimeter wherever possible and comply with applicable regulations.

Authorities should also seek to regulate and supervise entities that provide services related to GSCs, including those holding customer assets, such as custody service providers, to address operational, reputational, financial and market protection risks that may arise, including from the safeguarding of customer assets and private keys, and from the other activities that these entities could perform. Regulations and oversight should require the adequate safeguarding of customer assets and private keys, as well as appropriate risk disclosures and adequate protection of the users' ownership rights, including through prudent segregation and record-keeping requirements that minimise the risk of loss, misuse of or delayed access to assets. In considering appropriate rules and policies, authorities should also consider the proportionate risks that the service providers pose based on the size of and functions performed by the service providers.

Review of existing regulatory, supervisory and oversight requirements

Authorities should assess whether existing regulatory, supervisory and oversight requirements adequately address the risks of GSC functions and activities. Authorities should, if necessary or appropriate, clarify or supplement financial regulations if existing regulations may not adequately capture the risks of GSC functions and activities, and develop and implement regulations or approaches to address uncaptured risks.

Recommendation 3: Cross-border cooperation, coordination and information sharing

Authorities should cooperate and coordinate with each other, both domestically and internationally, to foster efficient and effective communication, information sharing and consultation in order to support each other in fulfilling their respective mandates and to ensure comprehensive regulation, supervision, and oversight of a GSC arrangement

across borders and sectors, and to encourage consistency of regulatory and supervisory outcomes.

Cooperation arrangements should be flexible, effective, efficient, inclusive, and multi-sectoral, and take into account the complexity and the potential evolution of the GSC arrangement and the risks it poses over time. They may take different forms (e.g. supervisory colleges, fora, networks, memoranda of understanding (MoUs), and ad-hoc arrangements). They should also consider the distinctive nature of GSC arrangements as usually consisting of multiple and oftentimes unrelated entities that interact and have varying roles and responsibilities.

Cooperation arrangements may be underpinned by bilateral and/or multilateral MoUs for cooperation and information sharing, and for crisis management and resolution, and complemented with mechanisms with a single focus, e.g. regarding AML/CFT or cyber security. These arrangements should also consider the potential need to seek cooperation from authorities in other jurisdictions to achieve regulatory objectives, e.g. in implementing recovery and resolution plans, or halting activities based in one jurisdiction having an adverse impact in another.

Authorities should consider a requirement that the arrangement takes into account the interests of each of the jurisdictions and sectors in which the GSC arrangements may be operating or seeking to operate, jurisdictions where the governance body, the providers of the GSC functions and activities and the GSC arrangement's users are located, where (spill over) risks reside, and the potentially differing impacts of GSC arrangements across jurisdictions and between advanced economies (AEs) and emerging market and developing economies (EMDEs).

In establishing cooperation arrangements, authorities should consider to take into account the "Key design considerations for establishing cooperation arrangements for GSCs" set out in Section 4.2.

Cooperation arrangements need to be effective in normal circumstances and should be adequately flexible to facilitate timely and effective management of crises, such as through enhanced cooperation and information sharing about GSC arrangements at times of crisis.

To foster effective cross-border cooperation and coordination, the FSB and SSBs will continue to promote consistency and a common understanding of key elements of regulatory, supervisory and oversight frameworks for GSCs.

Recommendation 4: Governance structures and decentralised operations

Authorities should require that GSC arrangements have in place and disclose a comprehensive governance framework with clear and direct lines of responsibility and accountability for all functions and activities within the GSC arrangement.

An adequate governance framework consistent with relevant international standards should be required for the entire network of GSC activities, functions and participants, given each part of the network can affect the other parts. GSC arrangements may vary in the degree of decentralisation of their governance design, but authorities should require that the ownership structure, governance and operation do not impede the effective application of relevant regulations and standards, consistent with the FSB High-level Recommendations. In particular,

authorities should require that GSC issuance be governed and operated by one or more identifiable and responsible legal entities or individuals ("governance body"). The governance structure should allow for timely human intervention, as and when needed or appropriate. Fully permissionless ledgers or similar mechanisms could pose particular challenges to the accountability and governance, and authorities should ensure that appropriate regulatory, supervisory and oversight requirements be effectively applied to such arrangements.

The governance structures and accountabilities should have a sound legal basis and be clear, transparent and disclosed to users, investors and other stakeholders. The governance body of a GSC should disclose how governance and accountability is allocated and how potential conflicts of interest are addressed among different entities within the GSC arrangement and in different jurisdictions, as well as clarify the limits of accountability and legal liability in any one jurisdiction. These disclosures should cover all functions and activities of the GSC arrangement, including but not limited to, setting rules and standards for participants of the arrangement, issuing stablecoins, operating the stabilisation mechanism, in particular the investing of the reserve assets as appropriate, providing the custody or trust services for reserve assets, and providing user-facing services such as exchanges and wallets.

Where a GSC arrangement relies on a third-party, including automated processes and algorithmic protocols (see recommendation 9 regarding the use of algorithmic protocols as a stabilisation mechanism), the GSC governance body should provide a comprehensive assessment and disclosure of how its reliance on the third-party does not impede its ability to meet regulatory requirements and expectations for performance, resilience, security, development and maintenance, and regulatory compliance.

Recommendation 5: Risk management

Authorities should require that GSC arrangements have effective risk management frameworks in place that comprehensively address all material risks associated with their functions and activities, especially with regard to operational resilience, cyber security safeguards and AML/CFT measures, as well as "fit and proper" requirements, if applicable, and consistent with jurisdictions' laws and regulations.

Authorities should require that GSC arrangements have in place policies that set out how all functions and activities within the GSC arrangement are subject to risk management measures that are appropriate to and commensurate with the specific risks that GSC arrangements may pose. If the risk from the fluctuation in the value of the underlying assets is borne, partially or totally by the GSC operator, the relevant prudential framework (e.g. market risk framework) should be applied to the GSC operator.

Authorities should require that GSC arrangements conduct proper due diligence (for example, by applying "fit and proper" standards, where applicable) into individuals involved in the management and control of the GSC arrangement, as well as those who exercise significant power or discharge significant responsibilities in relation to GSC activities.

Authorities should require that GSC arrangements have in place policies that comprehensively address all material risks associated with their functions and activities, such as operational risks (including fraud and cyber risks), compliance risk (including money laundering/terrorist financing

risks), and provide for appropriate consumer and investor protection, in line with legal obligations in jurisdictions where a GSC arrangement operates. Risk management measures and technical standards should cover relevant activities performed by providers of activities in the GSC arrangements, paying particular attention to compliance by permissionless or anonymous networks. Accordingly, authorities should ensure that GSC arrangements put appropriate AML/CFT measures in place consistent with FATF Standards, including requirements to comply with the FATF ”travel rule.”

Authorities should require that GSC arrangements conduct continuous risk assessments, contingency preparedness and continuity planning. Authorities should require that GSC arrangements conduct a robust assessment of how their technology model and the rules for transferring stablecoins or relevant assets provide assurance of settlement finality.

In addition to prudential requirements set forth in recommendation 9, authorities should require GSC arrangements to have comprehensive liquidity risk management practices and contingency funding plans that clearly set out the strategies and tools for addressing large number of redemptions i.e. run scenarios, and are regularly tested and operationally robust. The GSC arrangement should also have robust capabilities to measure, monitor and control funding and liquidity risks, including liquidity stress testing.

Recommendation 6: Data storage and access to data

Authorities should require that GSC arrangements have in place robust frameworks, including systems and processes for the collecting, storing, safeguarding and timely and accurate reporting of data. Authorities should have access to the data as necessary and appropriate to fulfil their regulatory, supervisory and oversight mandates.

GSC arrangements should implement and operate data management systems that record and safeguard in a discoverable format relevant data and information collected and produced in the course of their operations, while conforming to all applicable data privacy and other jurisdictional requirements. Adequate controls should be in place to safeguard the integrity and security of both on-chain and off-chain data and conform to applicable data protection regulation.

Authorities should be able to obtain timely and complete access to relevant data and information related to the GSC, wherever the data is located, to enable them to implement adequate regulatory, supervisory and oversight approaches, and recovery and resolution planning activities that capture the functions and activities of the GSC arrangement, in accordance with the level and nature of the risks posed.

Recommendation 7: Recovery and resolution of the GSC

Authorities should require that GSC arrangements have appropriate recovery and resolution plans.

Authorities should require that GSC arrangements have in place appropriate planning to support a recovery, resolution or orderly wind down under the applicable legal (or insolvency) frameworks, including continuity of any critical functions and activities within the GSC arrangement and prevent spillovers to the financial system.

Authorities should consider how such plans are implemented through effective contractual obligations among the entities in the GSC network and address the potential involvement of authorities in all of the jurisdictions in which the entities operate.

Recommendation 8: Disclosures

Authorities should require that GSC issuers and, where applicable, other participants in the GSC arrangements provide all users and relevant stakeholders with comprehensive and transparent information to understand the functioning of the GSC arrangement, including with respect to the governance framework, any conflicts of interest and their management, redemption rights, stabilisation mechanism, operations, risk management framework and financial condition.

Features of GSC arrangements that should be transparent to all users and relevant stakeholders include: the governance structure of the GSC arrangement; the allocation of roles and responsibilities assigned to operators or service providers within the GSC arrangement; the operation of the stabilisation mechanism; the composition of and investment mandate for the reserve assets (see Section 4.3 for common disclosure templates for reserve assets, which may be used by any stablecoin arrangement if there are no specific supervisory disclosure requirements applicable to the GSC); the custody arrangement and applicable segregation of reserve assets; available dispute resolution mechanisms or procedures for seeking redress or lodging complaints, as well as information on risk relevant for users.

Authorities should require that GSC issuers make appropriate disclosures to users and the market regarding the design of the stabilisation mechanism. Disclosures should also include details on the redemption rights of users and the redemption process.

The nature of these disclosures for all of the information (including the disclosure template in Section 4.3) will depend in part on the regulatory framework the issuer is operating under (e.g. whether the issuer is already subject to comprehensive supervision and regulation and disclosure requirements, e.g. as a bank and/or issuer of securities, etc.).

Information to be disclosed to users and relevant stakeholders should include the amount of GSC in circulation and the value and the composition of the assets in the reserve backing the GSC and should be subject to regular independent audits. Other information relevant to the functioning of the GSC arrangement, e.g. a list of available exchange platforms or wallet providers, should be made available on a regular basis, as appropriate.

Authorities should require GSC arrangements to have mechanisms to ensure the protection of the interests of users and counterparties, when a potential modification of the arrangement could have a material effect on the value, stability, or risk of the GSC.

Recommendation 9: Redemption rights, stabilisation, and prudential requirements

Authorities should require that GSC arrangements provide a robust legal claim to all users against the issuer and/or underlying reserve assets and guarantee timely redemption. For GSCs referenced to a single fiat currency, redemption should be at par

into fiat. To maintain a stable value at all times and mitigate the risks of runs, authorities should require GSC arrangements to have an effective stabilisation mechanism, clear redemption rights and meet prudential requirements.

Redemption rights

Authorities should require GSC arrangements to provide a robust legal claim and timely redemption to all users over a time period that is consistent with the treatment for other payment and settlement assets.[5] Redemption requests should be processed without undue costs for the user. In particular, the issuer should ensure that GSC users' redemption should not be unduly compromised by the disruption or failure of an intermediary or other relevant entity and infrastructure (e.g. through contractual or operational processes that allow for prompt redemption with the issuer directly in the event of an intermediary becoming unavailable).

The GSC arrangement should also provide adequate information on the process for redemption and the enforcement of GSC users' claims, and regarding how the GSC arrangement ensures smooth execution of the redemption process, including under stressed circumstances. GSC arrangements should not impose conditions that may unduly restrict the ability of GSC users to exercise their redemption rights (e.g. minimum thresholds). Any fees for redemption should be clearly communicated to users and should be proportionate, and not be high enough to become a de facto deterrent to redemption.

Stabilisation mechanism

Authorities should require GSCs to have an effective method to maintain a stable value at all times (a stabilisation method). An effective stabilisation method should include a reserve of assets that is at least equal to the amount of outstanding stablecoins in circulation at all times, unless the GSC issuer (or where applicable, the provider or providers of the issuance, redemption or stabilisation function of the GSC arrangement) is subject to adequate prudential requirements, oversight and safeguards equivalent to BCBS standards and delivers similar levels of protection to commercial bank money. A GSC should not rely on arbitrage activities to maintain a stable value at all times, and it should not derive its value from algorithms. Consequently, so-called algorithmic stablecoins do not meet recommendation 9 as they do not use an effective stabilisation method.

For GSCs that use a reserve-based stabilisation method ("reserve-based stablecoin"), authorities should ensure that there are robust requirements for the composition of reserve assets consisting only of conservative, high quality and highly liquid assets. Authorities should consider limitations to the reserve that would exclude speculative and volatile assets as well as assets where there is insufficient historical evidence and data of quality and liquidity, such as most crypto-assets. Due to the potential risk of fire sales of reserve assets, there should be particular attention to the nature, sufficiency and degree of risk-taking in terms of duration, credit quality, liquidity and concentration of a GSC's reserve assets. Reserve assets should be unencumbered and easily and immediately convertible into fiat currency at little or no loss of

[5] For stablecoins that are used for payments and are systemic, the PFMIs define timely as "as soon as possible, at a minimum by the end of day and ideally intraday."

value. The market value of reserve assets should meet or exceed the amount of outstanding claims or stablecoins in circulation at all times. In addition, risks of custodial arrangements for reserve assets should also be adequately managed and addressed. In particular, authorities should require reserve-based stablecoins to ensure safe custody and proper record-keeping of reserve assets and that ownership rights of reserve assets are protected at all times, including through segregation requirements from other assets of the GSC, members of its group and the custodian's assets. The reserve assets should be protected against claims of creditors of the GSC issuer, and members of the group of the GSC issuer, in particular in the event of insolvency of the issuer. GSC issuers should also consider the operational, credit and liquidity risks associated with the use of custody service providers to minimise the risk of loss, misuse of or delayed access to the reserve assets.

Prudential requirements

In order to have effective stabilisation methods, GSC arrangements should also be subject to appropriate prudential requirements (including capital and liquidity requirements) to provide that losses can be absorbed and there is sufficient liquidity to deal with outflows. Prudential requirements should take into account the risks of the reserve assets and operational risks (amongst other risks). Adequate capital buffers also contribute to maintaining confidence in the GSC and a stable value at all times. Such capital buffers should be consistent with the size of the GSC in circulation and proportionate to the risks of GSC arrangement.

Recommendation 10: Conformance with regulatory, supervisory and oversight requirements before commencing operations

Authorities should require that GSC arrangements meet all applicable regulatory, supervisory and oversight requirements of a particular jurisdiction before commencing any operations in that jurisdiction and adapt to new regulatory requirements as necessary and as appropriate.

Authorities should not permit the operation of a GSC arrangement in their jurisdiction unless the GSC arrangement meets all of their jurisdiction's regulatory, supervisory, and oversight requirements, including affirmative approval (e.g. licenses or registrations) where such a mechanism is in place. That includes requirements specific to crypto-assets as well as those general requirements (including consumer and investor protection regulation) that are applicable to the underlying economic and financial nature of crypto-assets, in line with the principle of "same activity, same risk, same regulation."

Where relevant, GSC arrangements should have the ability to adjust their operational features, processes and mechanisms as necessary or appropriate to maintain compliance with applicable regulatory requirements, consistent with international standards, if these evolve or change.

Before launching the arrangement and the provision of services to users in a particular jurisdiction, entities and individuals involved in the management and control of the GSC arrangement should understand applicable regulatory requirements. Where regulations of more than one jurisdiction may apply, understand which jurisdictions' rules are applicable to different aspects of the functions and activities performed and engage proactively with authorities.

Annex 1: Key design considerations for cooperation arrangements

There are different possible structures for cross-sectoral and cross-jurisdictional cooperation arrangements. The choice of structure should be tailored to the specific features of the stablecoin arrangement (e.g. risk profile, systemic footprint, level of interest of the relevant authorities). Those features may evolve or change over time, which may lead relevant authorities to consider whether and how the cooperation arrangement should be adapted. The following considerations are intended to inform a suitable structure for an effective cooperation arrangement while allowing for flexibility to accommodate the specificities of GSC.

Objective

The authorities that participate in the cooperation arrangement should have a common understanding of the objective of the arrangement.

The cooperation arrangement could be for information-sharing only or to facilitate coordinated decision-making or supervision on important matters such as authorisation or approval of material design changes, or to support the effective monitoring and enforcement of all applicable regulatory, supervisory and oversight requirements in each authority's jurisdiction, as contemplated in Recommendation 10.

Information-sharing arrangements could be beneficial if the objective of cooperation is to ensure all relevant authorities, on a cross-border and cross-sectoral basis have a common understanding of the functions, structure, operations, and risks of the GSC arrangement. These types of arrangements could be relevant for stablecoins which are not yet systemic but have the potential to become so at some stage in the future. Such structures would enable early involvement of authorities and facilitate decisions later that anticipate systemic potential.

Coordinating decision making could be explored through such cooperation arrangements for larger or more systemic GSCs where cooperation with respect to regulatory and supervisory decisions would be appropriate, and regulatory arbitrage is a key risk. For example, these arrangements could follow the spirit of PFMI Responsibility E which describes how authorities should cooperate with each other, both domestically and internationally, as appropriate and in a flexible manner, in promoting the safety and efficiency of FMIs.[6]

Structure and participation

Authorities should identify the structure and participation to achieve the stated objective of the cooperation.

The authorities involved in a cooperation arrangement should be the competent authorities that bear the responsibility for the regulation, supervision, recovery and resolution, and/or oversight of the stablecoin arrangement, as well as the central banks of issue for fiat-referenced GSCs.

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6 Cooperation arrangements between authorities in no way prejudice the statutory or legal or other powers of each participating authority, nor do these arrangements constrain in any way an authority's powers to fulfil its statutory or legislative mandate or its discretion to act in accordance with those powers.

Other relevant bodies (such as non-financial authorities) could also be involved, for information and consultation purposes. The competent authorities could be the ones responsible for one or more key functions of the GSC arrangements, such as governance, issuance, redemption, stabilisation, and the transfer functions. A cooperation arrangement should provide some flexibility to deal with scenarios where authorities face challenges in identifying competent and relevant authorities. Depending on the jurisdiction and/or type of licensing regime, the competent authorities could be a banking supervisor, securities or market regulator, or FMI/payment supervisor. Competent authorities can involve a combination of such authorities, as is the case already for traditional FMIs such as certain CCPs for example.

The cooperation arrangements could include different categories of membership considering: (i) the role and mandate of relevant authorities, including non-financial regulatory authorities; (ii) criteria to choose members of each category; (iii) considerations for inviting non-financial sector authorities; and (iv) ways to identify competent authorities in case no responsible entity is identified for one or more core function(s) of the GSC arrangement.

The footprint or materiality of the GSC across different jurisdictions, including when a GSC is under development or at an early stage, could impact the relevant authorities of the cooperation arrangement. There may also be differences in the level of activity for each jurisdiction the GSC arrangement operates in, including the possibility that the SC is not systemic in some of the jurisdictions with responsibility for the oversight and supervision of the GSC, while it could be systemically important in other jurisdictions. The membership should also consider the unique foreign exchange and capital account requirements facing emerging market jurisdictions if the GSC were to be widely used as a means of payment.

The possibility that the GSC will evolve or change, including by relocating core functions from one jurisdiction to another, may require a changing of the authorities with oversight and supervisory responsibilities. Given the ease with which GSC arrangements may change location of key functions, a cooperation arrangement should account for such a process in order to avoid any gaps in this respect.

Confidentiality and legal constraints

It is important that cross-jurisdictional cooperation is underpinned by effective information-sharing gateways and confidentiality and other safeguards to facilitate the exchange of information across borders and collective or combined risk assessment among the relevant supervisory and regulatory authorities. Information-sharing gateways and confidentiality and other safeguards could be established through the development and use of a multilateral information-sharing arrangement among authorities. Such arrangement could be informed by existing international standards and guidance, including the PFMI Responsibility E and the 2010 IOSCO Principles Regarding Cross-Border Supervisory Cooperation. Alternatively, members of the cooperation arrangement may rely on bilateral information-sharing arrangements between authorities. To the extent that the stablecoin activities fall within the existing regulatory perimeter authorities may be able to rely on existing information-sharing arrangements, which may not extend to other sectoral authorities or non-financial authorities but could be adapted to their needs.

Annex 2: Template for common disclosure of reserve assets

This template is designed as a common disclosure template for reserve-backed GSCs. This template is meant as a model for authorities seeking to supplement existing disclosure requirements or adopt new disclosure requirements. A common disclosure framework on the composition of and investment mandate for the reserve assets would help market participants consistently assess and compare the quality of the reserve portfolio and the GSC arrangement's ability to maintain redemption at par at all times. The template should not be interpreted as endorsing the inclusion of particular asset classes in the reserve portfolio, but only as illustrating the level of detail in which assets should be disclosed to users, investors and other stakeholders.

In addition to the categories listed below, the following should apply to all eligible assets included in the disclosure of reserve assets:

- Eligible assets must be unencumbered, meaning the assets are free of legal, regulatory, contractual, or other restrictions on the ability of the GSC arrangements to liquidate, sell, transfer, or assign the asset.
- Eligible assets should exclude unencumbered assets which the GSC arrangements does not have the operational capability to monetise to meet redemptions, including during periods of stress.
- Eligible assets should be under the control of the function charged with managing the reserve assets, meaning the function has the continuous authority, and legal and operational capability, to monetise any asset in the reserve.

GSC arrangements should report both the market value as of the reporting period as well as the daily average over the most recent quarter. The daily average should be presented as the simple average of daily observations.

	Illustrative asset categories[7]	Market value at month end	Weighted average maturity	Market Value (daily average over month-end)
1	Demand deposits at commercial banks			
2	Term deposits at commercial banks			
3	Money market fund shares			
4	Reverse repo (collateralized by government bills, bonds, or notes)			
5	Government bills			
6	Government bonds and notes			

[7] These asset categories are for illustrative purposes only. Relevant asset categories should be conservative, high quality, and highly liquid, consistent with the requirements in recommendation 9 and for reserve assets in each jurisdiction.

	Illustrative asset categories[7]	Market value at month end	Weighted average maturity	Market Value (daily average over month-end)
…	[…]			
7	Other assets[8]			
8	Of which, loans or extensions of credit to entities affiliated with the GSC			
9	Total assets of reserve portfolio			
10	Total stablecoins in circulation			

[8] Other assets should include information that is relevant for assessing and evaluating the quality of those assets, including for example the asset class, issuer, credit rating and maturity of each instrument.

Annex 3: Potential elements that could be used to determine whether a stablecoin qualifies as a GSC

A stablecoin's global systemic importance could be measured in terms of the impact that a stablecoin arrangement's financial or operational disruptions, or failure, can have on crypto-asset markets, the global financial system and the wider economy.

Given that a stablecoin may be used as a means of payment or store of value, and could be used in multiple jurisdictions, the criteria to be considered in determining whether a stablecoin qualifies as a GSC should take into account the potential uses in multiple jurisdictions.

The potential elements set out below that could be used to determine whether a stablecoin qualifies as a GSC build on the criteria that are often considered in determining the need for or degree of regulation, supervision, and oversight of FMIs (PFMI, 2012, also supplemented by guidance on the Application of the PFMI to stablecoin arrangements, issued in 2022), and global systemically important banks (BCBS, 2013). They include factors and considerations for authorities to determine whether a stablecoin has the potential to expand reach and adoption across multiple jurisdictions and has the potential to achieve substantial volume. Such potential elements are:

- Number and type of stablecoin users
- Number and value of transactions
- Size of reserve assets
- Value of stablecoins in circulation
- Market share in cross-border use in payments and remittances
- Number of jurisdictions with stablecoin users
- Market share in payments in each jurisdiction
- Redemption linked to a foreign currency or multiple currencies
- Interconnectedness with financial institutions and the broader economy,
- Interconnectedness with the wider crypto-assets ecosystem, other crypto-asset services and decentralised finance
- Integration with digital services or platforms (e.g. social networks, messaging applications)
- Available alternatives to using the GSC as a means of payment at short notice
- Business, structural and operational complexity

Glossary[9]

Algorithmic stablecoin

A stablecoin that purports to maintain a stable value via protocols that provide for the increase or decrease of the supply of the stablecoin in response to changes in demand.

Blockchain

A form of distributed ledger in which details of transactions are held in the ledger in the form of blocks of information. A block of new information is appended to the chain of pre-existing blocks via a computerised process by which transactions are validated.

Crypto-asset

A digital asset (issued by the private sector) that depends primarily on cryptography and distributed ledger or similar technology.

Crypto-asset ecosystem

The entire ecosystem that encompasses all crypto-asset activities, market and participants.

Crypto-asset issuer

An entity, person, or other structure that creates new crypto-assets.

Crypto-asset market

Any place or system that provides buyers and sellers the means to trade crypto-assets and the associated instruments, including lending, structured investment products, and derivatives. Crypto-asset markets facilitate the interaction between those who wish to offer and sell and those who wish to invest.

Crypto-asset services

Services relating to crypto-assets that may include, but are not limited to, distribution, placement, facilitating exchange between crypto-assets or against fiat currencies, custody, provisioning of non-custodial wallets, facilitating crypto-asset trading, borrowing or lending, and acting as a broker-dealer or investment adviser.

Crypto-asset service providers

Individuals and entities that provide crypto-asset services.

Crypto-asset activities

Activities serviced by a crypto-asset issuer or crypto-asset service provider.

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9 The glossary is for the purposes of this document and does not replace other existing taxonomies

Crypto-asset trading platform

Any platform where crypto-assets can be bought and sold, regardless of the platform's legal status.

Decentralised Finance (DeFi)

A set of alternative financial markets, products and systems that operate using crypto-assets and "smart contracts" (software) built using distributed ledger or similar technology.

DeFi protocols

A specialized autonomous system of rules that creates a program designed to perform financial functions.

Digital asset

A digital representation of value or contractual rights which can be used for payment or investment purposes.

Fiat-referenced stablecoin

A stablecoin that aims to maintain a stable value with reference to one or several fiat currencies and has the potential to be used as a means of payment and/or store of value.

Global stablecoin (GSC)

A stablecoin with an existing or potential reach and use across multiple jurisdictions and which could become systemically important in and across one or many jurisdictions, including as a means of making payments and/or store of value.

Reserve-based stablecoin

A stablecoin that aims to maintain a stable value with a dedicated pool of reserve assets.

Smart contract

Code deployed in a distributed ledger technology environment that is self-executing and can be used to automate the performance of agreement between entities. The execution of a smart contract is triggered when that smart contract is "called" by a transaction on the blockchain. If triggered, the smart contract will be executed through the blockchain's network of computers and will produce a change in the blockchain's "state" (for example, ownership of a crypto-asset will transfer between market participants).[10]

[10] There are unresolved questions regarding the legal status and enforceability of smart contracts.

Stablecoin

A crypto-asset that aims to maintain a stable value relative to a specified asset, or a pool or basket of assets.

Stablecoin arrangement

An arrangement that combines a range of functions (and related activities) that aims to maintain a stable value relative to a specified asset, or a pool or basket of assets. When discussing a stablecoin arrangement, reference is made to:

- Activity

  Typical activities in a stablecoin arrangement are: (i) establishing rules governing the stablecoin arrangement; (ii) issuing, creating and destroying stablecoins; (iii) managing reserve assets; (iv) providing custody/trust services for reserve assets; (v) operating the infrastructure; (vi) validating transactions; (vii) storing the private keys providing access to stablecoins (e.g. using a wallet); and (viii) exchanging, trading, reselling, and market making of stablecoins.

- Function

  Functions in a stablecoin arrangement are: (i) governing the arrangement; (ii) issuance, redemption and stabilisation of the value of coins; (iii) transfer of coins; and (iv) interaction with users for storing and exchanging coins.

- Governance body

  A body responsible for establishing and monitoring the rules governing the stablecoin arrangement which would cover, among other issues, the types of entities that could be involved in the arrangement, the protocol for validating transactions, and the manner in which the stablecoin is “stabilised”.

- Provider of function/activity

  An entity that provides a particular function or activity associated with that function in a stablecoin arrangement.

- User

  A person or entity that uses a stablecoin, e.g. for speculative trading, lending, borrowing, or as a means of payment or store of value.

- Validator node

  An entity that participates in the consensus mechanism in a distributed ledger or similar network. In the context of distributed ledger technology, a validator node will commit transaction blocks to the ledger once they are validated.

Wallet

An application or device for storing the cryptographic keys providing access to crypto-assets. A hot wallet is connected to the internet and usually takes the form of software for the user, while a cold wallet is a hardware that is not connected to the internet and stores the cryptographic keys.

Custodial wallet

A service in which crypto-assets are held by a service provider. A user interacts with the service provider to manage the user's crypto-assets. A custodial wallet is also known as a "hosted wallet".

Non-custodial wallet

Software or hardware that stores cryptographic keys for a user, making the user's crypto-assets accessible only to the user, and allowing the user to interact directly with the blockchain and the blockchain-based finance applications. A non-custodial wallet is also known as an "unhosted wallet".

17 July 2023

International regulation of crypto-asset activities: a proposed framework

Overview of responses to the consultation

1. Introduction

On 11 October 2022, the FSB published a proposed framework for the international regulation of crypto-asset activities. The core components of the framework were proposals for recommendations that promote the consistency and comprehensiveness of regulatory, supervisory and oversight approaches to crypto-asset activities and markets; and revised high-level recommendations for the regulation, supervision, and oversight of "global stablecoin" arrangements to address associated financial stability risks more effectively.

The FSB received 54 responses to the consultation which ended on December 15, 2022. Respondents included trade associations representing regulated financial institutions (14), crypto-asset companies (13), trade groups representing crypto-asset companies (9), regulated financial institutions (7), centres of research (3), individuals (2) and others (6).[1]

Overall, respondents mainly agree with the findings and conclusions presented in the two consultative reports, including the activity taxonomy, material risks and key regulatory challenges. There is also a broad recognition of the two sets of proposed recommendations. While respondents generally acknowledge that the FSB reports and the recommendations reflect the 'same activity, same risk, same regulation' principle, a few responses sought more clarity on details, with divergent directions due to the different perspectives and interests of different sectors. Some commenters requested clarifications for the intention of using different languages in areas common to both stablecoins and other crypto-assets, and asked for consistency to avoid confusion. On stablecoins, many respondents welcomed the revisions to the high-level recommendations for GSCs, including the disclosure template and considerations for cross-border cooperation. However, some commenters raised concerns with the broad definition of "could become GSC," while stablecoin issuers asked the FSB to clarify or explain its analysis of existing stablecoins, arguing their own stablecoins already meet the high-level recommendations.

On DeFi, many respondents asked to clarify whether DeFi is in scope of the high-level recommendations. Many crypto-asset respondents argued that DeFi is fundamentally different from other crypto-asset activities and should have a more bespoke regulatory framework or no regulation at all. On intermediaries, many commenters highlighted the failure of FTX and the need to strengthen aspects of the high-level recommendations, such as improving client asset

[1] Available here.

segregation, more detailed guidance on custody and wallet functions, and removing modifiers in the text on fundamental elements such as governance and risk management. However, other respondents highlighted the benefits of the combination of functions to offset risks (such as settlement) while requesting further work and analysis before determining which functions must be separated.

Having considered the feedback and suggestions received, some amendments and clarifications were made to the final reports and high-level recommendations, including the following:

- With regard to consistency between the two sets of recommendations, some parts of the report have been revised to align the texts in similar areas, however relevant differences in wording between the two sets of recommendations exist to account for the differences between stablecoins and crypto-assets.

- In light of the lessons learned from recent failures of crypto-asset service providers, the FSB has strengthened both sets of high-level recommendations in three areas:

  - **Safeguarding off client assets**. The high-level recommendations have been strengthened by stating that authorities should require that crypto-asset service providers maintain adequate safeguarding of customer assets and protect ownership rights, including in insolvency.

  - **Conflicts of interest.** The FSB has strengthened the high-level recommendations by stating that authorities should have in place requirements to address the risks associated with conflicts of interest.

  - **Cross-border cooperation.** The FSB's high-level recommendations have been strengthened on information sharing, including about the level of compliance of activities spanning multiple jurisdictions, especially those in jurisdictions that have not implemented international standards.

- The GSC report has been strengthened to require reserve-based stablecoins to ensure safe custody and proper record-keeping of reserve assets and that ownership rights of reserve assets are protected at all times.

- Finally, given crypto-assets and crypto-asset market structures continue to evolve rapidly, and the intended high-level and flexible design of the two sets of recommendations, the final report does not propose a more granular taxonomy of crypto-asset activities.

Set out below is a summary of the responses received to the 15 questions asked in the consultation.

2. Summary of feedback received

2.1. Scope and coverage of entities and risks (Q1,4,5)

While most respondents agreed that the FSB reports capture comprehensively the types of activities and market participants relevant for financial stability, there is a wide range of responses that expect more details on the taxonomy of crypto-assets and related participants. This includes a global terminology of types of crypto-assets, the classification of crypto-assets in terms of their functions, and differentiation of crypto-asset participants (especially intermediaries). On this basis, some respondents further requested clarification on what assets are and are not subject to the framework, such as CBDCs, NFTs, tokenised traditional financial assets. Banks are particularly keen on distinguishing bank-issued tokenised deposits from crypto-assets and suggest they be exempt from the FSB framework because they are already subject to prudential regulation. Most respondents did not propose a detailed taxonomy, but one suggested that an international definition can leverage from existing jurisdictional definitions, such as that provided by the MiCA.

Several respondents identified areas where further deep-dives or expanded scope may be considered. These areas include applicability of regulation to DeFi, accounting standards of crypto-assets, resolution of large market players, as well as areas that may not be regarded as the main focus on the FSB framework due to its financial stability mandate, such as issues related to AML/CFT, cyber security, consumer protection and market integrity, and even social and ethical risks. Some respondents suggested that regulation should be enhanced on custody services and market concentration. Commenters also argued that specific GSC thresholds are consistent and easier to operationalise.

A few respondents expressed concerns on regulatory fragmentation, with lack of harmonisation of regulatory standards. One respondent highlighted the insufficiency of binding forces of international rules, as the sanctioning power exercised on countries failing to conform to the international rules remain weak.

2.2. Structure of the two reports (Q2-3)

Many commenters suggested the FSB consider greater categorization and apply more graduated requirements based on granular taxonomies. Other commenters urged the FSB to reduce duplicity between the two sets of recommendations. A few responses suggested merging the recommendations, with additional requirements for GSCs if there is no visible benefit or considerations of keeping them separate. They noted that the finalised recommendations should at least be consistent with each other as much as possible. Most other responses related to the questions address more broadly the classification of crypto-assets.

As for the boundary between the scope of GSC and CA recommendations, a few respondents noted the need to narrow the potential discretion for national authorities to identify GSCs. As to the notion of ‘early readiness to regulate’ adopted by the GSC recommendations to reach potential GSCs, some responses showed objection to this, while one respondent suggested

complementing for the current framework by creating a regulatory category of 'non-GSC' stablecoins.

In additional to the classification of assets as summarised above, some responses called for a distinction between services provided by centralised entities and those conducted through DeFi protocols. Besides, there are recommendations on a distinction between assets issued by regulated institutions (such as tokenised deposits as raised above) and those issued by unregulated entities.

2.3. Crypto ecosystem characteristics, activity patterns, and risks (Q6-7)

Most responses agreed with the activity patterns and the associated risks identified by the FSB report. A few responses identified aspects that are worth further investigation with potential more nuanced analysis of risks and considerations of regulations, including:

- liquidity risk is more extensive than that related to Proof of Stake as listed in the Annex 1. They can arise in a number of other ways, such as through failure to maintain enough assets to meet redemptions;

- risks related to staking, oracle exploits, underlying infrastructure such as cross-chain bridges, systemic risks arising from market participants not subject to regulation (such as private wallets), risks from coding, risks related to DAOs and DApps, risks from failure to properly account for crypto liability in the relevant fiat currency;

While most respondents supported the 'same activity, same risk, same regulation' principle, some responses again debated on the extent to which the principle is applied to crypto-assets. Non-traditional crypto players seemed to be eager to stay away from the traditional regulatory approaches. In general, they argued that crypto-assets may not necessarily require the same regulations as traditional financial products because they present different risks even when performing the same activities. Furthermore, some crypto players urged for more recognition of the benefits brought by the new technology. A small number of respondents claimed that often labelled traditional risks may be offset by the 'benefits', such as traceability and transparency, faster settlement times, and automated smart contracts. One commentator stated that crypto intermediaries may appropriately combine functions and that conflicts of interest among intermediaries can be managed. They argued that traditional financial intermediary structures should not automatically apply to crypto. One respondent suggested risks are different between activities deployed in public chains and private chains.

Wallets and custody were identified by several respondents as areas that should be further investigated. However, suggestions on regulatory approaches vary, especially when focussing on unhosted wallets where respondents had divergent views of their risk features. One respondent stated that unhosted wallets should not be classified as an intermediary and are thus not subject to relevant regulation. One commentator, in contrast, called for stronger language on unhosted wallets related to risk management and suggested tracking the language from the GSC consultation. Another respondent noted that unhosted wallets have different functions, some of which have asset-recovery abilities. In this regard, more nuanced discussion and differentiated regulation may be considered in light of their different functions.

One other respondent focused on the interaction with wallets of DeFi protocols and suggested further analysis is needed to understand the entity who has access to the keys. Contrary to the views from the crypto industry, one bank respondent believed that crypto custody regulation should be considerably strengthened by mirroring the bank custody practices and standards to ensure adequate customer protections.

In addition, DeFi is also identified by some respondents as an area that needs broader discussion (this is referred to in responses to other questions too).

2.4. Regulatory challenges and gaps (Q8)

Most respondents agreed that the FSB report correctly identified the key regulatory and supervisory challenges. Almost all responses to this question are either the reiteration of the importance of key areas that the FSB report has already included, or suggestions on further work by the FSB.

The urgency to promote cross-border regulatory coordination is again raised by a number of respondents. They noted the lack of the harmonisation of regulatory frameworks and supervisory practices among different jurisdictions especially those non-FSB member jurisdictions falling outside of the implementation scope, which may not only give rise to regulatory fragmentation, but also add to uncertainty of regulatory outcomes. One respondent pointed out the need of reciprocal and mutual recognition of regulation between national authorities.

Other areas that receive attention from respondents include intermediaries combing functions, DeFi and technical risks.

Specifically, as for the intermediaries combining functions, the responses came from two centralised crypto exchanges. These responses strived to push against the potential application of common traditional regulation. In particular, the crypto exchange respondents expressed concern of mandatory segregation of functions implied in CA recommendation 9. They indicated that despite the existence of conflicts of interest between certain functions, they may be effectively managed by appropriate asset segregation, supported by internal systems and controls and external audit. One centralised exchange argued that they can facilitate real-time settlement and reduce the number of intermediaries involved in a transaction, thus reducing transaction costs. The respondent believed that adoption of separate governance and management will ensure keeping decisions independent to mitigate conflicts of interest.

A few respondents raised DeFi as an area with significant regulatory challenges. One respondent is of view that the building blocks of FSB recommendations, including governance, accountability and other compliance requirements, are also relevant to DeFi but are not practical nor appropriate to be applied directly given the different nature of DeFi. One bank respondent, while also noting the challenges of regulating DeFi, expressed scepticism of relying on the issuer as regulatory anchor point because issuers themselves confront the same challenges as regulators, such as the absence of legal entities for ecosystem players and no legal instruments available to impose any requirements. They further indicated that to only focus on issuers will lead to disadvantages for regulated traditional institutions. One

respondent specifically mentioned that some DeFi protocols are operating as FMIs and are not adequately regulated.

Respondents also drew attention to the technical risks of crypto-assets. One respondent introduced the various efforts of the technical communities and companies to protect assets of unhosted wallets from technical risks, including scams, phishing, and cyber-attacks, and noted technical risks relating to the quality of software or a technical product are not, and should not be, addressed through regulation. Another respondent highlighted the importance of code auditing and the need to set expectations, for example, about the frequency with which the codebase for platforms is audited and the expected content and conduct of such audits.

Some other responses include: the need for CDBCs to be covered by the recommendations, the need to consider the potential of GSCs to enhance financial stability for crypto-asset markets by allowing for sophisticated market participants engaging in crypto-asset markets to efficiently shift and rebalance capital across global markets, as well as the potential for innovation across the traditional financial and crypto markets to help solve supervisory and oversight challenges. A group response from lawyers noted the need for a separate recommendation for having in place appropriate mechanisms for consumer and investor protection.

2.5. CA recommendations (Q9-10)

Respondents generally supported the differentiated requirements between issuers and service providers. However, there is a strong appeal from the crypto respondents not to introduce too prescriptive requirements. Several of them proposed the notion of ‘voluntary adoption’ of risk management and data disclosure, and argued against applying risk-based regulation to crypto industry. Crypto industry respondents are also dissatisfied with disclosure requirements. One commenter proposed that the FSB and national authorities should leverage blockchain data to increase transparency in crypto-asset activities. Several respondents advised international standard setters to help with further work, including detailed taxonomy and how to apply regulation to DeFi, or respond to activities where the issuer is not traceable or identifiable.

In contrast, some traditional institutions, especially banks, stressed that risk management and data standards for crypto-asset activities should not be lower than those in traditional finance. For example, one bank respondent suggested that authorities should adopt third-party risk management requirements for non-bank crypto-asset service providers consistent with those that apply to banks. One other respondent stressed that policymakers should ensure that data protection and cybersecurity requirements are applied consistently to those providing crypto-asset related services between banks and non-bank service providers.

Some respondents again referred to the need for a granular taxonomy as a basis of differentiated treatment of different activities or market participants. One respondent suggested the distinction. However, there are very few details or concrete suggestions on how to further differentiate activities or entities. At a high-level, a few respondents noted the need to differentiate between DeFi and other crypto-asset activities (or between DeFi and CeFi), which is again consistent with responses to other questions. Specifically, one respondent suggested that trading through a centralised exchange is very different from interacting unhosted wallets with DeFi protocols usually in Automated Market Maker (AMM) mode.

A few respondents urged for more international consistency given the borderless nature of crypto-assets. One respondent noted that existence of crypto disclosure requirement in some jurisdictions (e.g., the EU's white paper requirements under MiCA) but there is lack of standardised disclosure requirements globally.

2.6. Stablecoin developments (Q11)

Respondents were generally supportive of the analysis of recent market developments and existing stablecoins. Some commenters emphasised the need to update the analysis in light of recent market events and various market developments. While other commenters requested the analysis provide more differentiation between various types of stablecoins. A few commenters requested the FSB provide the rationale for stating "most existing stablecoin arrangements do not meet the FSB's high-level recommendations." These commenters stressed the difference in reserve assets and stabilisation mechanisms (e.g., stablecoins backed by fiat currencies, stablecoins backed by crypto-assets, and algorithmic stablecoins) would lead to different risks. Another commenter noted the different risks of stablecoins backed by crypto-assets affiliated with the promoter (e.g., Terra) and those backed by crypto-assets unaffiliated with the promoter (e.g., DAI). Commenters associated with stablecoin issuers noted their compliance programs and claimed they meet the FSB high-level recommendations.

2.7. GSC Recommendations (Q12-15)

The majority of respondents agreed with the recommendations put forward in the report. Some commenters asked the FSB to clarify or strengthen certain recommendations. Some of the points raised may warrant further consideration by CWG members. For example, a few respondents emphasised the need for segregation requirements on reserve assets to ensure reserve assets are available to coinholders if needed (with one respondent suggesting further guidance on dealing with insolvency). Similarly, a few respondents suggested the FSB should strengthen recommendation 2 with respect to custody services, notably concerning the segregation of client assets and clients' ownership rights. Two respondents requested additional clarity about the regulatory responsibilities of each function involved in a GSC. A few respondents also noted that the risks stemming from the combination of functions should be more explicitly covered in the GSC recommendations. And a couple of respondents noted FSB recommendations could include broader third-party risk management requirements.

With respect to governance arrangements and algorithmic stablecoins, a few respondents suggested more flexibility, while others supported a more cautious approach by authorities as suggested in the recommendations. For instance, whilst one respondent noted that Recommendation 9 "effectively bans" algorithmic stablecoins, another one would instead welcome "a more explicit ban" of these stablecoins due to significant risks of deviation from the peg. With respect to governance, a couple of respondents challenged the need to have in place a comprehensive governance body applying to "all functions of the GSC arrangement" (with one respondent noting this could lead to the pooling of risks), but another respondent noted the importance of ensuring accountability for managing all functions and activities within a stablecoin arrangement. One respondent also called for the FSB to permit a tiered redemption model, where GSC issuers would only be held accountable for redemptions of

customers of the issuer, provided all other token holders have “reasonable avenues for timely redemption” through other providers.

In addition, a number of respondents noted the need for clarifications or further guidance in areas that may be particularly relevant for SSBs and international organisations to consider (and where international policy coordination would be useful). For instance, many respondents encouraged further consideration and/or clarification of the treatment of tokenized traditional assets, and, in particular, tokenized commercial bank deposits, which are already subject to existing regulation. Some respondents noted bank tokenized deposits should be more explicitly distinguished from GSC and should not be captured by the CA or the GSC recommendations. A few respondents emphasised the need for clarifications on the treatment of unhosted wallets and peer-to-peer crypto-asset transactions, including on alignment with the FATF travel rule. Additionally, one respondent emphasised the need for further clarification from the FSB or the International Accounting Standards Board (IASB) around accounting for crypto-assets, while two respondents noted code auditing standards would also be helpful.

A few respondents were of the view that the recommendations for CAs and for GSCs involved some duplication and required harmonisation of language and coverage. One of the respondents recommended a few ways some of the recommendations that apply to GSCs could be extended to cover all crypto-assets.

About half of the respondents either agreed with the proposed disclosure template for reserve-backed GSC or did not make specific comments. A number of respondents provided suggestions as to how the disclosure template could be designed. Some examples are set out below:

- mirror the template to e-money and money market mutual funds or use the existing format on collateral reporting by custodians;
- disclosure requirements should be proportionate to the potential risk they represented;
- disclosures are made public akin to a prospectus;
- expand the scope of application of disclosure template to other non-global stablecoins; and
- further data points would be needed for specific types of stablecoins, depending on the use case and composition of the reserve.

Some respondents also proposed that more granular information could be included in the disclosure template, such as distinctions between types of asset classes, haircuts, name and location of accounts, mechanisms to determine values, among other details.

Several respondents sought clarification on how crypto-assets held by the GSC should be disclosed; whether wallet addresses of relevant holdings of need to be disclosed; whether the template covers partially reserve backed stablecoins; and whether “daily average over month” should be used instead of "daily average over month-end” for the breakdown of market value.

On a related note, respondents sought clarification and elaborations in relation to FSB high-level recommendation 8 of the regulation, supervision and oversight of GSC arrangements. Examples include clarifying who the "relevant stakeholders" are in respect of a GSC arrangement, and elaborating what is meant by "mechanism to ensure protection of interests of users and counterparties"; and the framework for audit and assurance of reserve assets.

About one third of the responses explicitly address Question 15 of the consultation, which sought feedback on the elements that could be used to determine whether a stablecoin qualifies as a GSC (annex 4). Respondents provided suggestions in two broad areas. First, the FSB should provide more clarity on how the suggested criteria will be applied in practice. In some cases, respondents asked for quantitative thresholds used to define GSC; and second, the FSB should narrow down the number of criteria used to define GSC, focusing in particular on the use of stablecoins for cross-border payments and interconnections with the financial system. In general, respondents did not provide any evidence to support their suggestions.